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Access authorization for audit information in a multi-tenancy data management system

Abstract

Methods, systems, and devices for data management are described. A data management system (DMS) may provide backup services for multiple tenants. The DMS may receive a request to provide a first user of the DMS with audit information associated with an entity of the DMS. The DMS may identify context information for a log-in session associated with the request. The context information may include an identifier (ID) of a tenant associated with the request. The DMS may identify, based on the ID of the tenant, authorization information associated with the tenant. The authorization information may indicate that the tenant has access to a set of entities within a hierarchy associated with the DMS. The DMS may determine whether to output the requested audit information to the first user based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

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Background/Summary

FIELD OF TECHNOLOGY

(1) The present disclosure relates generally to data management, including techniques for access authorization for audit information in a multi-tenancy data management system (DMS).

BACKGROUND

(2) A data management system (DMS) may be employed to manage data associated with one or more computing systems. The data may be generated, stored, or otherwise used by the one or more computing systems, examples of which may include servers, databases, virtual machines, cloud computing systems, file systems (e.g., network-attached storage (NAS) systems), or other data storage or processing systems. The DMS may provide data backup, data recovery, data classification, or other types of data management services for data of the one or more computing systems. Improved data management may offer improved performance with respect to reliability, speed, efficiency, scalability, security, or ease-of-use, among other possible aspects of performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of a computing environment that supports access authorization for audit information in a multi-tenancy data management system (DMS) in accordance with aspects of the present disclosure.

(2) FIG. 2 illustrates an example of a multi-tenancy system that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.

(3) FIG. 3 illustrates an example of a computing object hierarchy that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.

(4) FIG. 4 illustrates an example of an authentication hierarchy that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.

- (5) FIG. 5 illustrates an example of a computing environment that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.
- (6) FIG. 6 illustrates an example of a process flow that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.
- (7) FIG. 7 illustrates an example of a process flow that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.
- (8) FIGS. 8 and 9 illustrate block diagrams of devices that support access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.
- (9) FIG. 10 illustrates a block diagram of an audit authorization component that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.
- (10) FIG. 11 illustrates a diagram of a system including a device that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.
- (11) FIGS. 12 through 14 illustrate flowcharts showing methods that support access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(12) A multi-tenancy data management system (DMS) may provide backup services for multiple tenants (e.g., organizations or business units). A multi-tenancy DMS also may have resources across cloud platforms and on-premise data centers. In multi-tenant scenarios, multiple tenants (e.g., organizations or business units) may share data management resources. Some multi-tenant scenarios may be multi-level, with multiple hierarchical levels of tenants. For example, resources of a backup and recovery system may be shared among multiple higher-level tenants, and at least some of the higher-level tenants may be associated with one or more levels of lower-level tenants (e.g., subtenants), with resources associated with a higher-level tenant being shared by multiple subtenants of that tenant.

(13) As one such example, which may be referred to as an enterprise scenario, an information technology (IT) services unit of a business (e.g., of a corporation) may be a tenant of a DMS, and multiple other business units of the same business (e.g., within the same corporation) may be subtenants of the IT services unit, and accordingly, may share the same data management services. As another such example, some tenants of a DMS may be multi-service providers (MSPs). An MSP may be a higher-level tenant of a backup and recovery system and may provide IT and data management services to multiple distinct customers, which may be separate businesses that are subtenants of the MSPs. For example, the MSP may subscribe to data management services and resources from the DMS, and the MSP may use those services and resources to in turn provide data management service to the MSP's subtenants (e.g., an MSP subtenant may not directly subscribe to the DMS, such as due to a lack of internal expertise in configuring or managing the resources or services of the DMS, and thus the MSP subtenant may instead be customer of the MSP, which may directly subscribe to the DMS and use the MSP's subscription to offer data management services to the MSP subtenant).

(14) There may be many tenants of the DMS, and some or all of the tenants may have any number of subtenants. The tenants of the DMS may be enterprise tenants, MSP tenants, other types of entities, or any combination thereof. Further, an entity that is a subtenant of a higher-level tenant may itself have one or more subtenants. That is, there may be three or more levels of tenants—in general, any quantity of levels may exist.

(15) In some examples, a DMS may obtain and store audit information for users of the system. For example, the DMS may generate audit information when a user logs in to the DMS and may update the audit information to indicate operations (e.g., audit events) performed by the user when logged in to the DMS or otherwise associated with the user.

(16) Techniques, systems, and devices described herein provide for a DMS to determine whether a first user that requests access to audit information for an entity of a multi-tenant system has

permission to access the audit information associated with the entity, while accounting for impacts of a multi-level hierarchical multi-tenancy environment. The first user may be, for example, an administrator of a tenant that wants to review operations performed by the second user. In some examples, if the entity is a second user, the DMS may determine whether to grant the first user access to the audit information associated with the second user based on an authentication domain that is assigned to a tenant to which the first user has access. For example, the DMS may assign authentication domains to one or more tenants of the multi-tenant system. An authentication domain may include one or more subtenants, user groups, or users that the tenant has authorization over. If the second user is within an authentication domain that has been assigned to the tenant associated with the first user (e.g., the tenant has authorization over the second user), the first user may have permission to access the audit information for the second user. The second user may be a subject of the audit information (e.g., an entity that performed the audit events) or an object of the audit information (e.g., an entity that was the target of the audit events), or both. The DMS may determine an identifier (ID) of the tenant to which the first user has access based on the context for a log-in session for the first user. Thus, the DMS may provide the first user access to the audit information if the tenant of the first user has authorization over the second user, and the DMS may refrain from providing the audit information otherwise.

(17) Additionally, or alternatively, if the entity is a computing object of the DMS, the DMS may determine whether to grant the first user access to the audit information based on a set of computing objects to which the tenant of the first user has access within a computing object hierarchy of the DMS. For example, if the first user requests to view audit information associated with audit events performed on one or more computing objects or other entities, the DMS may output the requested audit information to the first user if the one or more computing objects are included in the set of computing objects to which the tenant of the user has access, and the DMS may refrain from outputting the requested audit information otherwise. The DMS may thereby perform access control for audit information based on a subject of the audit information (e.g., a user), or an object of the audit information (e.g., a user, a computing object, or some other entity), or both.

(18) In some examples, when generating an audit report for a given user, the DMS may store an ID of a tenant associated with the user in the audit information. The DMS may use this tenant information to filter audit reports by tenants, which may improve techniques for tenants to manage subtenants and corresponding users. For example, a higher-level tenant may request that the DMS display all stored audit information for users of one or more subtenants. The DMS may scan a database of previously obtained audit information and select the audit information that includes the tenant ID of the subtenant(s). The DMS may output the requested audit information for the subtenant after filtering the audit information database. The described techniques may thereby provide for the DMS to retrieve and provide audit information with improved reliability, security, and efficiency.

(19) FIG. 1 illustrates an example of a computing environment **100** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The computing environment **100** may include a computing system **105**, a DMS **110**, and one or more computing devices **115**, which may be in communication with one another via a network **120**. The computing system **105** may generate, store, process, modify, or otherwise use associated data, and the DMS **110** may provide one or more data management services for the computing system **105**. For example, the DMS **110** may provide a data backup service, a data recovery service, a data classification service, a data transfer or replication service, one or more other data management services, or any combination thereof for data associated with the computing system **105**.

(20) The network **120** may allow the one or more computing devices **115**, the computing system **105**, and the DMS **110** to communicate (e.g., exchange information) with one another. The network

120 may include aspects of one or more wired networks (e.g., the Internet), one or more wireless networks (e.g., cellular networks), or any combination thereof. The network **120** may include aspects of one or more public networks or private networks, as well as secured or unsecured networks, or any combination thereof. The network **120** also may include any quantity of communications links and any quantity of hubs, bridges, routers, switches, ports or other physical or logical network components.

(21) A computing device **115** may be used to input information to or receive information from the computing system **105**, the DMS **110**, or both. For example, a user of the computing device **115** may provide user inputs via the computing device **115**, which may result in commands, data, or any combination thereof being communicated via the network **120** to the computing system **105**, the DMS **110**, or both. Additionally or alternatively, a computing device **115** may output (e.g., display) data or other information received from the computing system **105**, the DMS **110**, or both. A user of a computing device **115** may, for example, use the computing device **115** to interact with one or more user interfaces (e.g., graphical user interfaces (GUIs)) to operate or otherwise interact with the computing system **105**, the DMS **110**, or both. Though one computing device **115** is shown in FIG. 1, it is to be understood that the computing environment **100** may include any quantity of computing devices **115**.

(22) A computing device **115** may be a stationary device (e.g., a desktop computer or access point) or a mobile device (e.g., a laptop computer, tablet computer, or cellular phone). In some examples, a computing device **115** may be a commercial computing device, such as a server or collection of servers. And in some examples, a computing device **115** may be a virtual device (e.g., a virtual machine). Though shown as a separate device in the example computing environment of FIG. 1, it is to be understood that in some cases a computing device **115** may be included in (e.g., may be a component of) the computing system **105** or the DMS **110**.

(23) The computing system **105** may include one or more servers **125** and may provide (e.g., to the one or more computing devices **115**) local or remote access to applications, databases, or files stored within the computing system **105**. The computing system **105** may further include one or more data storage devices **130**. Though one server **125** and one data storage device **130** are shown in FIG. 1, it is to be understood that the computing system **105** may include any quantity of servers **125** and any quantity of data storage devices **130**, which may be in communication with one another and collectively perform one or more functions ascribed herein to the server **125** and data storage device **130**.

(24) A data storage device **130** may include one or more hardware storage devices operable to store data, such as one or more hard disk drives (HDDs), magnetic tape drives, solid-state drives (SSDs), storage area network (SAN) storage devices, or network-attached storage (NAS) devices. In some cases, a data storage device **130** may comprise a tiered data storage infrastructure (or a portion of a tiered data storage infrastructure). A tiered data storage infrastructure may allow for the movement of data across different tiers of the data storage infrastructure between higher-cost, higher-performance storage devices (e.g., SSDs and HDDs) and relatively lower-cost, lower-performance storage devices (e.g., magnetic tape drives). In some examples, a data storage device **130** may be a database (e.g., a relational database), and a server **125** may host (e.g., provide a database management system for) the database.

(25) A server **125** may allow a client (e.g., a computing device **115**) to download information or files (e.g., executable, text, application, audio, image, or video files) from the computing system **105**, to upload such information or files to the computing system **105**, or to perform a search query related to particular information stored by the computing system **105**. In some examples, a server **125** may act as an application server or a file server. In general, a server **125** may refer to one or more hardware devices that act as the host in a client-server relationship or a software process that shares a resource with or performs work for one or more clients.

(26) A server **125** may include a network interface **140**, processor **145**, memory **150**, disk **155**, and

computing system manager **160**. The network interface **140** may enable the server **125** to connect to and exchange information via the network **120** (e.g., using one or more network protocols). The network interface **140** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. The processor **145** may execute computer-readable instructions stored in the memory **150** in order to cause the server **125** to perform functions ascribed herein to the server **125**. The processor **145** may include one or more processing units, such as one or more central processing units (CPUs), one or more graphics processing units (GPUs), or any combination thereof. The memory **150** may comprise one or more types of memory (e.g., random access memory (RAM), static random access memory (SRAM), dynamic random access memory (DRAM), read-only memory ((ROM), electrically erasable programmable read-only memory (EEPROM), Flash, etc.). Disk **155** may include one or more HDDs, one or more SSDs, or any combination thereof. Memory **150** and disk **155** may comprise hardware storage devices. The computing system manager **160** may manage the computing system **105** or aspects thereof (e.g., based on instructions stored in the memory **150** and executed by the processor **145**) to perform functions ascribed herein to the computing system **105**. In some examples, the network interface **140**, processor **145**, memory **150**, and disk **155** may be included in a hardware layer of a server **125**, and the computing system manager **160**) may be included in a software layer of the server **125**. In some cases, the computing system manager **160** may be distributed across (e.g., implemented by) multiple servers **125** within the computing system **105**.

(27) In some examples, the computing system **105** or aspects thereof may be implemented within one or more cloud computing environments, which may alternatively be referred to as cloud environments. Cloud computing may refer to Internet-based computing, wherein shared resources, software, and/or information may be provided to one or more computing devices on-demand via the Internet. A cloud environment may be provided by a cloud platform, where the cloud platform may include physical hardware components (e.g., servers) and software components (e.g., operating system) that implement the cloud environment. A cloud environment may implement the computing system **105** or aspects thereof through Software-as-a-Service (SaaS) or Infrastructure-as-a-Service (IaaS) services provided by the cloud environment. SaaS may refer to a software distribution model in which applications are hosted by a service provider and made available to one or more client devices over a network (e.g., to one or more computing devices **115** over the network **120**). IaaS may refer to a service in which physical computing resources are used to instantiate one or more virtual machines, the resources of which are made available to one or more client devices over a network (e.g., to one or more computing devices **115** over the network **120**).

(28) In some examples, the computing system **105** or aspects thereof may implement or be implemented by one or more virtual machines. The one or more virtual machines may run various applications, such as a database server, an application server, or a web server. For example, a server **125** may be used to host (e.g., create, manage) one or more virtual machines, and the computing system manager **160** may manage a virtualized infrastructure within the computing system **105** and perform management operations associated with the virtualized infrastructure. The computing system manager **160** may manage the provisioning of virtual machines running within the virtualized infrastructure and provide an interface to a computing device **115** interacting with the virtualized infrastructure. For example, the computing system manager **160** may be or include a hypervisor and may perform various virtual machine-related tasks, such as cloning virtual machines, creating new virtual machines, monitoring the state of virtual machines, moving virtual machines between physical hosts for load balancing purposes, and facilitating backups of virtual machines. In some examples, the virtual machines, the hypervisor, or both, may virtualize and make available resources of the disk **155**, the memory, the processor **145**, the network interface **140**, the data storage device **130**, or any combination thereof in support of running the various applications. Storage resources (e.g., the disk **155**, the memory **150**, or the data storage device **130**) that are virtualized may be accessed by applications as a virtual disk.

(29) The DMS **110** may provide one or more data management services for data associated with the computing system **105** and may include DMS manager **190** and any quantity of storage nodes **185**. The DMS manager **190** may manage operation of the DMS **110**, including the storage nodes **185**. Though illustrated as a separate entity within the DMS **110**, the DMS manager **190** may in some cases be implemented (e.g., as a software application) by one or more of the storage nodes **185**. In some examples, the storage nodes **185** may be included in a hardware layer of the DMS **110**, and the DMS manager **190** may be included in a software layer of the DMS **110**. In the example illustrated in FIG. **1**, the DMS **110** is separate from the computing system **105** but in communication with the computing system **105** via the network **120**. It is to be understood, however, that in some examples at least some aspects of the DMS **110** may be located within computing system **105**. For example, one or more servers **125**, one or more data storage devices **130**, and at least some aspects of the DMS **110** may be implemented within the same cloud environment or within the same data center.

(30) Storage nodes **185** of the DMS **110** may include respective network interfaces **165**, processors **170**, memories **175**, and disks **180**. The network interfaces **165** may enable the storage nodes **185** to connect to one another, to the network **120**, or both. A network interface **165** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. The processor **170** of a storage node **185** may execute computer-readable instructions stored in the memory **175** of the storage node **185** in order to cause the storage node **185** to perform processes described herein as performed by the storage node **185**. A processor **170** may include one or more processing units, such as one or more CPUs, one or more GPUs, or any combination thereof. The memory **150** may comprise one or more types of memory (e.g., RAM, SRAM, DRAM, ROM, EEPROM, Flash, etc.). A disk **180** may include one or more HDDs, one or more SSDs, or any combination thereof. Memories **175** and disks **180** may comprise hardware storage devices. Collectively, the storage nodes **185** may in some cases be referred to as a storage cluster or as a cluster of storage nodes **185**.

(31) The DMS **110** may provide a backup and recovery service for the computing system **105**. For example, the DMS **110** may manage the extraction and storage of snapshots **135** associated with different point-in-time versions of one or more target data sources within the computing system **105**. A snapshot **135** of a data source (e.g., a virtual machine, a database, a filesystem, a virtual disk, a virtual desktop, or other type of computing system or storage system) may be a file (or set of files) that represents a state of the data source (e.g., the data thereof) as of a particular point in time. A snapshot **135** may also be used to restore (e.g., recover) the corresponding data source as of the particular point in time corresponding to the snapshot **135**. A data source of which a snapshot **135** may be generated may be referred to as snappable. Snapshots **135** may be generated at different times (e.g., periodically or on some other scheduled or configured basis) in order to represent the state of the computing system **105** or aspects thereof as of those different times. In some examples, a snapshot **135** may include metadata that defines a state of the data source as of a particular point in time. For example, a snapshot **135** may include metadata associated with (e.g., that defines a state of) some or all data blocks included in (e.g., stored by or otherwise included in) the data source. Snapshots **135** (e.g., collectively) may capture changes in the data blocks over time. Snapshots **135** generated for the target data sources within the computing system **105** may be stored in one or more storage locations (e.g., the disk **155**, memory **150**, the data storage device **130**) of the computing system **105**, in the alternative or in addition to being stored within the DMS **110**, as described below.

(32) To obtain a snapshot **135** of a target data source associated with the computing system **105** (e.g., of the entirety of the computing system **105** or some portion thereof, such as one or more databases, virtual machines, or filesystems within the computing system **105**), the DMS manager **190** may transmit a snapshot request to the computing system manager **160**. In response to the snapshot request, the computing system manager **160** may set the target data source into a frozen

state (e.g., a read-only state). Setting the target data source into a frozen state may allow a point-in-time snapshot **135** of the target data source to be stored or transferred.

(33) In some examples, the computing system **105** may generate the snapshot **135** based on the frozen state of the data source. For example, the computing system **105** may execute an agent of the DMS **110** (e.g., the agent may be software installed at and executed by one or more servers **125**), and the agent may cause the computing system **105** to generate the snapshot **135** and transfer the snapshot to the DMS **110** in response to the request from the DMS **110**. In some examples, the computing system manager **160** may cause the computing system **105** to transfer, to the DMS **110**, data that represents the frozen state of the target data source, and the DMS **110** may generate a snapshot **135** of the target data source based on the corresponding data received from the computing system **105**.

(34) Once the DMS **110** receives, generates, or otherwise obtains a snapshot **135**, the DMS **110** may store the snapshot **135** at one or more of the storage nodes **185**. The DMS **110** may store a snapshot **135** at multiple storage nodes **185**, for example, for improved reliability. Additionally or alternatively, snapshots **135** may be stored in some other location connected with the network **120**. For example, the DMS **110** may store more recent snapshots **135** at the storage nodes **185**, and the DMS **110** may transfer less recent snapshots **135** via the network **120** to a cloud environment (which may include or be separate from the computing system **105**) for storage at the cloud environment, a magnetic tape storage device, or another storage system separate from the DMS **110**.

(35) Updates made to a target data source that has been set into a frozen state may be written by the computing system **105** to a separate file (e.g., an update file) or other entity within the computing system **105** while the target data source is in the frozen state. After the snapshot **135** (or associated data) of the target data source has been transferred to the DMS **110**, the computing system manager **160** may release the target data source from the frozen state, and any corresponding updates written to the separate file or other entity may be merged into the target data source.

(36) In response to a restore command (e.g., from a computing device **115** or the computing system **105**), the DMS **110** may restore a target version (e.g., corresponding to a particular point in time) of a data source based on a corresponding snapshot **135** of the data source. In some examples, the corresponding snapshot **135** may be used to restore the target version based on data of the data source as stored at the computing system **105** (e.g., based on information included in the corresponding snapshot **135** and other information stored at the computing system **105**, the data source may be restored to its state as of the particular point in time). Additionally or alternatively, the corresponding snapshot **135** may be used to restore the data of the target version based on data of the data source as included in one or more backup copies of the data source (e.g., file-level backup copies or image-level backup copies). Such backup copies of the data source may be generated in conjunction with or according to a separate schedule than the snapshots **135**. For example, the target version of the data source may be restored based on the information in a snapshot **135** and based on information included in a backup copy of the target object generated prior to the time corresponding to the target version. Backup copies of the data source may be stored at the DMS **110** (e.g., in the storage nodes **185**) or in some other location connected with the network **120** (e.g., in a cloud environment, which in some cases may be separate from the computing system **105**).

(37) In some examples, the DMS **110** may restore the target version of the data source and transfer the data of the restored data source to the computing system **105**. And in some examples, the DMS **110** may transfer one or more snapshots **135** to the computing system **105**, and restoration of the target version of the data source may occur at the computing system **105** (e.g., as managed by an agent of the DMS **110**, where the agent may be installed and operate at the computing system **105**).

(38) In response to a mount command (e.g., from a computing device **115** or the computing system **105**), the DMS **110** may instantiate data associated with a point-in-time version of a data source

based on a snapshot **135** corresponding to the data source (e.g., along with data included in a backup copy of the data source) and the point-in-time. The DMS **110** may then allow the computing system **105** to read or modify the instantiated data (e.g., without transferring the instantiated data to the computing system). In some examples, the DMS **110** may instantiate (e.g., virtually mount) some or all of the data associated with the point-in-time version of the data source for access by the computing system **105**, the DMS **110**, or the computing device **115**.

(39) In some examples, the DMS **110** may store different types of snapshots, including for the same data source. For example, the DMS **110** may store both base snapshots **135** and incremental snapshots **135**. A base snapshot **135** may represent the entirety of the state of the corresponding data source as of a point in time corresponding to the base snapshot **135**. An incremental snapshot **135** may represent the changes to the state—which may be referred to as the delta—of the corresponding data source that have occurred between an earlier or later point in time corresponding to another snapshot **135** (e.g., another base snapshot **135** or incremental snapshot **135**) of the data source and the incremental snapshot **135**. In some cases, some incremental snapshots **135** may be forward-incremental snapshots **135** and other incremental snapshots **135** may be reverse-incremental snapshots **135**. To generate a full snapshot **135** of a data source using a forward-incremental snapshot **135**, the information of the forward-incremental snapshot **135** may be combined with (e.g., applied to) the information of an earlier base snapshot **135** of the data source along with the information of any intervening forward-incremental snapshots **135**, where the earlier base snapshot **135** may include a base snapshot **135** and one or more reverse-incremental or forward-incremental snapshots **135**. To generate a full snapshot **135** of a data source using a reverse-incremental snapshot **135**, the information of the reverse-incremental snapshot **135** may be combined with (e.g., applied to) the information of a later base snapshot **135** of the data source along with the information of any intervening reverse-incremental snapshots **135**.

(40) In some examples, the DMS **110** may provide a data classification service, a malware detection service, a data transfer or replication service, backup verification service, or any combination thereof, among other possible data management services for data associated with the computing system **105**. For example, the DMS **110** may analyze data included in one or more data sources of the computing system **105**, metadata for one or more data sources of the computing system **105**, or any combination thereof, and based on such analysis, the DMS **110** may identify locations within the computing system **105** that include data of one or more target data types (e.g., sensitive data, such as data subject to privacy regulations or otherwise of particular interest) and output related information (e.g., for display to a user via a computing device **115**). Additionally or alternatively, the DMS **110** may detect whether aspects of the computing system **105** have been impacted by malware (e.g., ransomware). Additionally or alternatively, the DMS **110** may relocate data or create copies of data based on using one or more snapshots **135** to restore the associated data source within its original location or at a new location (e.g., a new location within a different computing system **105**). Additionally or alternatively, the DMS **110** may analyze backup data to ensure that the underlying data (e.g., user data or metadata) has not been corrupted. The DMS **110** may perform such data classification, malware detection, data transfer or replication, or backup verification, for example, based on data included in snapshots **135** or backup copies of the computing system **105**, rather than live contents of the computing system **105**, which may beneficially avoid adversely affecting (e.g., infecting, loading, etc.) the computing system **105**.

(41) In some examples, the DMS **110** may be a multi-tenancy DMS **110** that may provide backup services for multiple tenants (e.g., organizations or business units). The tenants may be organized into a hierarchy of tenants. For example, resources of the DMS **110** may be shared among multiple higher-level tenants, and at least some of the higher-level tenants may be associated with one or more levels of lower-level tenants (e.g., subtenants), with resources associated with a higher-level tenant being shared by multiple subtenants of that tenant. The DMS **110** may obtain and store audit information for users of the system. For example, the DMS **110** may generate audit information

when a user logs in to the DMS **110** and may update the audit information to indicate operations (e.g., audit events) performed by the user when logged in to the DMS **110** or otherwise associated with the user. The DMS **110** may store the audit information in a storage node **185** of the DMS **110**, or some other location in the computing environment **100**.

(42) Techniques, systems, and devices described herein provide for the DMS **110** or a central manager of the DMS **110** to determine whether a first user that requests access to audit information for a second user of a multi-tenant system has permission to access the audit information associated with the second user, while accounting for impacts of a multi-level hierarchical multi-tenancy environment. The first user may be, for example, an administrator of a tenant that wants to review operations performed by the second user. The DMS **110** may determine whether to grant the first user access to the audit information based on an authentication domain that is assigned to a tenant to which the first user has access. For example, the DMS **110** may assign authentication domains to one or more tenants of the multi-tenant system. An authentication domain may include one or more subtenants, groups, or users that the tenant has authorization over. If the second user is within an authentication domain that has been assigned to the tenant associated with the first user (e.g., the tenant has authorization over the second user), the first user may have permission to access the audit information for the second user. The DMS **110** may determine an ID of the tenant to which the first user has access based on the context for a log-in session for the first user. Thus, the DMS **110** may provide the first user access to the audit information if the tenant of the first user has authorization over the second user, and the DMS **110** may refrain from providing the audit information otherwise.

(43) In some examples, when generating an audit report for a given user, the DMS **110** may store an ID of a tenant associated with the user in the audit information at a storage node **185** or other storage location. The DMS **110** may use this tenant information to filter audit reports by tenants, which may improve techniques for tenants to manage subtenants and corresponding users. For example, a higher-level tenant may request that the DMS **110** display all stored audit information for users of a subtenant. The DMS **110** may scan a database of previously obtained audit information and select the audit information that includes the tenant ID of the subtenant. The DMS **110** may output the requested audit information for the subtenant after filtering the audit information database. The described techniques may thereby provide for the DMS **110** to retrieve and provide audit information with improved reliability, security, and efficiency.

(44) FIG. 2 illustrates an example of a multi-tenancy system **200** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The multi-tenancy system **200** may implement or be implemented by aspects of the computing environment **100** described with reference to FIG. 1. For example, a DMS **110** may provide backup and recovery protection for data sources for multiple tenants and/or subtenants.

(45) As described herein, a global organization (e.g., a tenant **205**) may provide IT services, including backup and recovery protection via a DMS **110**, to multiple tenants (e.g., tenant **210-a** and tenant **210-b**). Additionally, each tenant may further have subtenants. For example, the tenant **210-a** may have a subtenant **215-a** and a subtenant **215-b**. For example, the tenant **205** may be the IT services unit of an organization, and the tenant **210-a** and the tenant **210-b** may be business units of or teams within the organization. The subtenant **215-a** and the subtenant **215-b** may be sub-business units or sub-teams of the business unit corresponding to the tenant **210-a** (e.g., working groups within the business unit). The subtenant **215-c** similarly may be a sub-business unit or sub-teams of the business unit corresponding to the tenant **210-b**. As another example, the tenant **205** may be an MSP, and the tenant **210-a** and the tenant **210-b** may be different enterprises/customers (e.g., organizations) of the MSP. The subtenant **215-a**, the subtenant **215-b**, and the subtenant **215-c** may be business units and/or working groups/entities/teams of the enterprises/customers corresponding to the tenant **210-a** and the tenant **210-b**.

(46) In some examples, the tenant **205** corresponds to a DMS **110** that controls backup and

recovery resources that are used to provide backup and recovery protection to the various tenants **210-a** and subtenants **215** of the organization. An administrative user of the tenant **205** may access the DMS **110** to configure and allocate resources (e.g., computing objects) that are used to support backup and recovery for data sources associated with the various tenants and subtenants. For example, the user may access a user interface of the DMS **110** to create the tenants **210-a** and **210-b** and to assign the respective backup and recovery resources to the created tenants **210-a** and **210-b**. Assignment of resources to a tenant may include updating metadata (e.g., RBAC metadata) associated with the respective resources to indicate respective tenant or subtenant assignments. In some cases, the administrative user may assign, to a tenant or subtenant using the user interface of the DMS **110**, a data source that is to be backed-up using a respective resource, a backup or recovery procedure that may be performed using the respective resource, and/or a storage capacity for the backup and recovery resource. Assignment of a data source, procedure, or capacity may include updating the metadata (e.g., RBAC metadata) associated with the backup and recovery resource (e.g., computing object) that is to be used by the tenant or subtenant.

(47) In some cases, the administrative user may access the user interface of the DMS **110** to assign users to the tenants **210** or subtenants **215**. For example, the administrative user of the tenant **205** may assign a second administrative user to the tenant **210-a** such that the second administrative user may access the platform for backup and recovery management, as well as further subtenant creation and resource assignment, data source assignment, procedure assignment, and capacity assignment. A third demonstrative user may be similarly assigned to the tenant **210-b**. User assignment may be restricted or controlled based on hierarchical techniques, as described herein with respect to computing object assignment.

(48) As described herein, users may access a user interface associated with the DMS **110** to control various backup and recovery aspects related to a tenant **205** or **210** or subtenant **215**. In some examples, the user interface may be supported by a platform or application that is used to manage multiple DMSs **110**, multiple tenants **205**, subtenants **215**, etc. In some examples, an authorized user may access the platform or application to control backup and recovery procedures, as well as tenant or subtenant creation and assignment. Each tenant **205** or **210** or subtenant **215** may be associated with a “context” of the platform or application. An application context refers to a state of an application that allows a user to manage to control aspects of backup and recovery associated with a particular tenant **205** or **210** or subtenant **215**. Thus, a user may access an application context associated with the tenant **210-a** and the user may view resources, procedures, etc. that are assigned to the tenant **210-a** as well as create subtenants of the tenant **210-a** (e.g. subtenants **215-a** and **215-b**) and assign subsets of resources to the created subtenants. Thus, when discussing a user accessing a user interface of the DMS **110** herein, the user may access the application context associated with a tenant or subtenant to perform various functions and procedures described herein.

(49) As described herein, the DMS **110** may provide for an RBAC scheme such that users associated with each tenant/subtenant may access only the computing objects assigned to the given tenant/subtenant. Accordingly, the tenants **210** and subtenants **215** may share a single DMS **110** and/or a single data management cluster without unauthorized access by any tenant **210**) or subtenant **215** to computing objects or files assigned to a different tenant **210** or subtenant **215**. For example, one business unit of an enterprise may not access computing objects or files assigned to a different business unit of the enterprise. As another example, one customer of an MSP may not access computing objects or files assigned to a different customer of the MSP.

(50) In accordance with aspects of the present disclosure, the DMS **110** may receive, at a user interface associated with the DMS **110**, a request, by a user of a tenant **210** of the DMS **110**, to access audit information associated with another entity of the DMS **110**. The other entity may be a subject of the audit information (e.g., an entity that performed the audit event, such as a user) or an object of the audit information (e.g., an entity that is the target of the audit event, such as a user or a computing object, or some other entity). The DMS **110** may determine whether the user that

requests access to the audit information for an entity of the multi-tenancy system **200** has permission to access the audit information associated with the entity, while accounting for impacts of a multi-level hierarchical multi-tenancy system **200**. The user may be, for example, an administrator of a tenant **210** that wants to review operations performed by a second user or operations performed on a second user or a set of one or more computing objects.

(51) In some examples, the DMS **110** may determine whether to grant the user access to the audit information based on an authentication domain that is assigned to a tenant **210** to which the user has access. For example, the DMS **110** may assign authentication domains to one or more tenants **210** of the multi-tenancy system **200**. An authentication domain may include one or more subtenants **215**, user groups, or users that the tenant **210** has authorization over, as described in further detail elsewhere herein, including with reference to FIG. 4. If the user requests to view audit information associated with a second user, and the second user is within an authentication domain that has been assigned to the tenant (**210**) associated with the user, such as the tenant **210-a**, the user may have permission to access the audit information for the second user. The DMS **110** may determine an ID of the tenant **210-a** to which the user has access based on the context for a log-in session for the user. In some examples, the DMS **110** may determine whether to grant the user access to the audit information based on a set of computing objects to which the tenant **210** to which the user has access is assigned or authorized to access, as described in further detail elsewhere herein, including with reference to FIG. 3. Thus, the DMS **110** may provide the user access to the audit information if the tenant **210** of the user has authorization over or access to the entity of the requested audit information, and the DMS **110** may refrain from providing the audit information otherwise.

(52) In some examples, when generating an audit report for a given user, the DMS **110** may store an ID of a tenant associated with the user in the audit information at a storage node or other storage location. The DMS **110** may use this tenant information to filter audit reports by tenants **210** or subtenants **215**, which may improve techniques for tenants to manage subtenants **215** and corresponding users. For example, a higher-level tenant **210**, such as the tenant **210-a**, may request that the DMS **110** display all stored audit information for users of a subtenant **215**, such as the subtenant **215-a**. The DMS **110** may scan a database of previously obtained audit information and select the audit information that includes the tenant ID of the subtenant **215-a**. The DMS **110** may output the requested audit information for the subtenant **215-a** after filtering the audit information database. In some examples, the user of the tenant **210-a** may request to view all audit information for a group or set of two or more subtenants **215** of the tenant **210-a**, and the DMS **110** may filter the report data accordingly. The described techniques may thereby provide for the DMS **110** to retrieve and provide audit information with improved reliability, security, and efficiency.

(53) FIG. 3 illustrates an example of computing object hierarchy **300** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The computing object hierarchy **300** may implement or be implemented by aspects of the computing environment **100** described with reference to FIG. 1. For example, a DMS **110** may provide backup and recovery protection for data sources for multiple tenants and/or subtenants via one or more data management clusters. For example, FIG. 3 illustrates a first data management cluster **315-a** and a second data management cluster **315-b**, which may provide protection for data sources associated with a first tenant **305-a**, a second tenant **305-b**, and a third tenant **305-c**.

(54) Each of the first data management cluster **315-a** and the second data management cluster **315-b** may include a set of computing objects (e.g., resources such as virtual machines or databases) which may be organized according to a hierarchical relationship. For example, the first data management cluster **315-a** may include the computing object **310-a**, which has as descendants the computing object **310-b** and the computing object **310-e**. The computing object **310-b** has as descendants the computing object **310-c** and the computing object **310-d**, and the computing object

310-d further has as a descendant the computing object **310-g**. The computing object **310-e** has as a descendent the computing object **310-f**.

(55) The second data management cluster **315-b** may include the computing object **310-h**, which has as descendants the computing object **310-i** and the computing object **310-l**. The computing object **310-i** has as a descendant the computing object **310-j**, and the computing object **310-j** further has as a descendant the computing object **310-k**. The computing object **310-l** has as a descendent the computing object **310-m**.

(56) As described herein, the multiple tenants (the first tenant **305-a**, the second tenant **305-b**, and the third tenant **305-c**) may share data management resources. More specifically, multiple tenants of a DMS **110** may share computing objects **310** of a same data management clusters **315**. For example, the first tenant **305-a** and the second tenant **305-b** may both be assigned computing objects **310** within the first data management cluster **315-a**, and the first tenant **305-a** and the third tenant **305-c** may both be assigned computing objects **310** within the second data management cluster **315-b**. The assignment of computing objects **310** of the data management clusters **315** may respect the hierarchical relationship among the computing objects **310**. For example, assignment of a top-level computing object such as the computing object **310-b** to the first tenant **305-a** may result in assignment of (e.g., an implicit assignment of) the computing object **310-c**, the computing object **310-d**, and the computing object **310-g** to the first tenant **305-a**, as the computing object **310-c**, the computing object **310-d**, and the computing object **310-g** are descendants of the computing object **310-b** within the computing object hierarchy of the first data management cluster **315-a**. Similarly, assignment of the computing object **310-e** to the second tenant **305-b** may result in assignment of the computing object **310-f** to the second tenant **305-b**. As another example, assignment of the computing object **310-i** to the first tenant **305-a** may result in assignment of the computing object **310-j** and the computing object **310-k** to the first tenant **305-a**. As another example, assignment of the computing object **310-l** to the third tenant **305-c** may result in assignment of the computing object **310-m** to the third tenant **305-c**. Such implicit assignment of resources based on the computing object hierarchy may simplify management of the DMS **110** for an administrator (e.g., an IT services unit or an MSP). For example, an MSP may assign a full data management cluster **315** to a tenant to achieve assignment of all backup computing objects from that data management cluster **315** to the tenant. As another example, an MSP may assign a Vcenter from a data management cluster **315** to assign all of the virtual machines from that Vcenter to the tenant.

(57) As described herein, the DMS **110** may provide for a multi-tenancy RBAC scheme such that users associated with each tenant/subtenant may access only the computing objects assigned to the given tenant/subtenant. Multi-tenancy RBAC supports in-depth computing object level access control granularity for data management systems such as the DMS **110** with multiple data management clusters. Multi-tenancy RBAC supports both authorizing a full data management cluster **315** to a tenant (e.g., assigning all of the computing objects **310** of a given data management cluster **315** to a given tenant), and assignment of specific computing objects **310** of a data management cluster **315** to a given tenant. A user associated with the first tenant **305-a** may not access computing objects assigned to the second tenant **305-b** or the third tenant **305-c**, a user associated with the second tenant **305-b** may not access computing objects assigned to the first tenant **305-a** or the third tenant **305-c**, and a user associated with the third tenant **305-c** may not access computing objects assigned to the first tenant **305-a** or the second tenant **305-b**. For example, a tenant dashboard (e.g., a user interface view for a tenant account at a computing device **115**) may show relevant statistics and information regarding authorized computing objects **310** for that tenant, and the tenant may only manage data backup for the authorized computing objects **310**. The multi-tenancy RBAC may prevent information leakage across tenants via enforcing access control at all user interfaces, events, audits, reports, etc., for a tenant, such that a tenant is not able to view or access direct or aggregated information about computing objects **310** that are not

assigned to that tenant. A tenant may not bypass the access control enforcement in either the control plane or the cluster side via federated login.

(58) As an example, an MSP account with multiple cloud accounts or on-premise data management clusters (e.g., the first data management cluster **315-a** may be a first cloud account or on-premise data management cluster of the MSP and the second data management cluster **315-b** may be a second cloud account or on-premise data management cluster of the MSP), and the MSP may assign a specific computing object **310** within the multiple cloud accounts or on-premise data management clusters of the MSP to a given tenant (e.g., customer) of the MSP. Further, multi-tenancy RBAC may authorize different permissions (e.g., read-only, read-write) on different computing objects **310**. For example, a tenant (e.g., the first tenant **305-a**) may have read-only access to a virtual machine (e.g., the computing object **310-c**) but read-write access to a MSSQL database (e.g., the computing object **310-d**).

(59) Techniques, systems, and devices described herein provide for the DMS **110** to determine whether a user has permission to access audit information associated with a given entity based on the tenant **305** associated with a given log-in session by the user. As described in further detail elsewhere herein, including with reference to FIG. 5, a DMS **110** may obtain and store audit information to track an audit event. The audit event may correspond to an operation performed by a user of the DMS **110**. The user that performs the operation may be referred to as a subject of the audit event and the entity on which the operation is being performed may be referred to as an object of the audit event, in some examples described herein. For example, if a user (UserA) updates one or more settings associated with a computing object **310** (e.g., a vCenter) of the DMS **110**, the DMS **110** may store audit information that includes a record of the operations performed by the user (e.g., the subject) on the computing object **310** (e.g., the object). If a first user (UserA) resets one or more configuration settings (e.g., a multi-factor authentication (MFA)) for a second user (UserB) of the DMS **110**, the DMS **110** may store audit information that includes a record of the operations performed by the first user (e.g., the subject) on the second user (e.g., the object). Access control for audit information may be performed, by the DMS **110**, based on the subject of an audit event, an object of the audit event, or both.

(60) In some examples, if the access control is performed at least in part based on an object of an audit event, and the object is a computing object **310**, the DMS **110** may utilize the computing object hierarchy **300** to facilitate the access control. For example, if the DMS **110** receives a request to access audit information associated with audit events performed on one or more computing objects **310**, the DMS **110** may determine context information associated with a log-in session of a user that sent the request. The context information may indicate a tenant ID of a tenant **305** to which the user is logged in. The DMS **110** may determine authorization information that indicates a relationship between the tenant ID of the tenant **305** and the one or more computing objects **310** associated with the requested audit information. If the one or more computing objects **310** are included in or below a set of computing objects **310** to which the tenant **305** associated with the request has access in the computing object hierarchy **300**, the DMS **110** may determine to output the audit information to the user in response to the request. The DMS **110** may otherwise refrain from outputting the audit information to the user if the one or more computing objects **310** are not included in or below the set of computing objects **310** to which the tenant **305** associated with the request has access in the computing object hierarchy **300**.

(61) For example, if the user is signed in to the tenant **305-a** and requests to view audit information associated with the computing objects **310-e** and **310-f**, the DMS **110** may refrain from outputting the requested audit information to the user, because the requested computing objects **310-e** and **310-f** may not be included in the set of computing objects **310-b**, **310-c**, **310-d**, and **310-g** to which the tenant **305-a** has access in the computing objects hierarchy **300**. If the user is signed into the tenant **305-a** and requests to view audit information associated with the computing objects **310-b** and **310-c**, the DMS **110** may output the requested audit information to the user based on

determining that the tenant **305-a** is authorized to access the computing objects **310-b** and **310-c**.

(62) The DMS **110** may thereby utilize the computing object hierarchy **300** to determine whether a user is authorized to access audit information associated with an audit event performed on one or more computing objects **310** (e.g., the object of the audit event). In some examples, the DMS **110** may perform access control based on the subject of audit information, or both the subject and the object, as described in further detail elsewhere herein, including with reference to FIGS. **4-6**. The described techniques may thereby provide for the DMS **110** to retrieve and provide audit information with improved reliability, security, and efficiency.

(63) FIG. **4** illustrates an example of an authentication hierarchy **400** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The authentication hierarchy **400** may implement or be implemented by aspects of the computing environment **100**, the multi-tenancy system **200**, or the computing object hierarchy **300**, as described with reference to FIGS. **1-3**. For example, the authentication hierarchy **400** indicates a hierarchy of authorization for access to users **420** and user groups **415** (e.g., groups of one or more users, tenants, or tenant organizations) within a multi-tenancy system. The multi-tenancy system may represent a system of multiple tenants supported by a DMS **110**, and the multiple tenants may be organized within a tenant hierarchy, as described with reference to FIG. **2**.

(64) A user **420** of the DMS **110** may represent an administrator or other client user that logs in to the DMS **110** via a user interface. The user **420** may log in to the DMS **110** through a given tenant **405**. For example, an employee of a company may represent an example of a user **420**, and the company may represent an example of a tenant **405**. The employee may log in to the DMS **110** via a user interface associated with the company. In some examples described herein, a user **420** may be referred to as an entity of the DMS **110**.

(65) The tenants **405** in the multi-tenancy system may obtain authority to access and manage one or more users **420** of the DMS. If a tenant **405** has authorization over a user **420** as described herein, the tenant **405** may be able to monitor and adjust one or more settings or parameters associated with the user **420**. For example, a tenant **405** may be able to authorize a change of username and password for a user **420** to which the tenant **405** has access, or the tenant **405** may be able to monitor audit information associated with a user **420** to which the tenant **405** has access, or both. After a tenant **405** gains access to or authorization over a certain user **420**, the tenant **405** may grant the user **420** with access to the tenant **405**. For example, the tenant **405** may authorize the user **420** to perform actions inside the tenant **405** (e.g., via the log-in request of the user **420**).

(66) The authentication hierarchy **400** may represent a hierarchy of authorizations for tenants **405** to access and manage users **420**. The nodes in the authentication hierarchy **400** may include users **420**, user groups **415** of two or more users **420** (e.g., tenants **405** or subtenants), authentication domain roots **425**, and authentication domains **410**. If a tenant **405** is assigned to or has access to a node in the authentication hierarchy **400**, the tenant **405** also has access to all nodes below that node in the authentication hierarchy **400**. An authentication domain root **425** may represent a root of the authentication hierarchy (e.g., a highest node in the hierarchy). In some examples, a global organization (e.g., an MSP) may have access to the authentication domain root **425**, and may thereby automatically have authorization to access all descendances (e.g., other nodes, such as authentication domains **410**, users **420**, user groups **415**) below the authentication domain root **425**. The authentication domain root **425** may be subdivided into one or more authentication domains **410**. The assignment of different tenants **405** to different nodes in the authentication hierarchy **400** may be managed and performed by the DMS **110**, one or more higher level tenants **405** or global organizations, or both.

(67) A tenant **405** within the multi-tenancy system may have access to or may be assigned to one or more different authentication domains **410**. As used herein, an authentication domain **410** may represent a domain (e.g., a single sign-on (SSO) domain, or some other type of domain), of users **420** that follow the same authentication methods (e.g., password-based authentication, SSO,

lightweight directory access protocol (LDAP), or any other type of authentication method). A tenant **405** may have authorization over all users within a given authentication domain **410**. Additionally, or alternatively, the tenant **405** may be provided authorization over a single user **420** or a user group **415**. A user group **415** may represent a group of one or more users **420**.

(68) In the example of FIG. 4, a first tenant **405-a** (Tenant 1) may have access to the authentication domain **410-a**. The first tenant **405-a** may thereby have access to each node below the authentication domain **410-a** in the authentication hierarchy **400**. For example, the first tenant **405-a** may have access to the user **420-a** and the user group **415**, which may include at least the user **420-b**. The user **420-a** and the user group **415** may be within a same domain (e.g., an SSO domain or some other type of domain), which may be linked to the authentication domain **410-a**. Thus, the first tenant **405-a** may have access to all users in the domain.

(69) A second tenant **405-b** (Tenant 2) may have access to the authentication domain **410-b**. The second tenant **405-b** may thereby have access to each node below the authentication domain **410-b** in the authentication hierarchy **400**. In the example of FIG. 4, the authentication domain **410-b** may include a single user **420-c**. Thus, the second tenant **405-b** may have access to the user **420-c**.

(70) In some examples, a tenant **405** may have access directly to a single user **420** or to a single user group **415** (e.g., user groups **415-a** or **415-b**). For example, the first tenant **405-a** may have access to the authentication domain **410-a** and at least two other nodes in the authentication hierarchy **400**. The first tenant **405-a** may have access to the user **420-d** and the user group **415-b**. The first tenant **405-a** may thereby have direct authorization over the user **420-d** and the user group **415-b** without an authorization domain. The user **420-e** may be assigned to or below the user group **415-b** in the authentication hierarchy. Accordingly, the first tenant **405-a** may automatically have access to the user **420-e** based on being assigned to the user group **415-b**.

(71) Although the tenants **405-a** and **405-b** are shown as having access to different authentication domains **410** in FIG. 4, it is to be understood that, in some cases, more than one tenant **405** may be assigned to a same authentication domain **410**. The authentication hierarchy **400** may include any quantity of nodes and any quantity of levels. For example, a third tenant (not pictured in FIG. 4) may have access to just the user **420-b** or to another authentication domain below the authentication domain **410-a**, or both.

(72) The authentication hierarchy **400** may be different than a hierarchy of tenants **405** within the multi-tenancy system, such as the hierarchy illustrated in the multi-tenancy system **200** described with reference to FIG. 2. For example, even if the first tenant **405-a** is higher than the second tenant **405-b** in the tenant hierarchy, the first tenant **405-a** may not necessarily or automatically have access to users **420** of the second tenant **405-b** that is assigned to or lower than the first tenant **405-a** in the tenant hierarchy. Additionally, or alternatively, the second tenant **405-b** may not necessarily or automatically have access to users **420** of the first tenant **405-a** that is higher than the second tenant **405-b** in the tenant hierarchy. Rather, access authorization may be determined based on which nodes the tenants **405** are assigned to in the authentication hierarchy **400**. In some aspects, one or more objects or other entities may be included in the authentication hierarchy **400**.

(73) Techniques, systems, and devices described herein provide for a DMS **110** to utilize the authentication hierarchy **400**, as well as tenant information stored with audit information at the DMS **110**, to authorize access requests for authorization information. If the DMS **110** receives a request for a first user **420** to access audit information associated with a second user **420**, the DMS **110** may determine whether the first user **420** is authorized to access the second user **420** before outputting the audit information, which may improve security and reliability in the multi-tenancy system. The access control for audit information may be performed based on a subject or object of an audit event. That is, the second user **420** may be the subject of the audit event (e.g., the performer) or the object of the audit event (e.g., the target of the audit event).

(74) In the example of FIG. 4, the first tenant **405-a** may be authorized to access audit information of both the user **420-a** and the user **420-b**. In some examples, the user **420-a** may be logged in to

the DMS **110** through the first tenant **405-a** (e.g., the user **420-a** may have authorization to access the first tenant **405-a**) and the user **420-b** may be logged in to the DMS **110** through a different tenant **405** (e.g., the user **420-b** may have authorization to access a third tenant **405** not pictured in FIG. **4**). If the user **420-a** is logged in to the DMS **110** through the first tenant **405-a** and outputs a request to the DMS **110** to access audit information for the user **420-b** and the user group **415-b**, the DMS **110** may determine that the user **420-a** is authorized to access the audit information for the user **420-b** and the user group **415-b** and may output the requested audit information accordingly. For example, the DMS **110** may determine, based on context information for the log-in session associated with the user **420-a**, that the user **420-a** is associated with the first tenant **405-a**. The DMS **110** may subsequently scan the authentication hierarchy **400** to determine a hierarchical relationship, in the authentication hierarchy **400**, between the first tenant **405-a** and the requested entities, including the user **420-b** and the user group **415-b**. The hierarchical relationship may indicate that the user **420-b** is below an authentication domain **410-a** to which the first tenant **405-a** is assigned and the user group **415-b** is assigned to the first tenant **405-a** in the authentication hierarchy **400**. The DMS **110** may output the audit information accordingly.

(75) If the user **420-a** requests to access audit information associated with the user **420-c**, the DMS **110** may determine that the tenant **405-a** is not authorized to access the user **420-c** or the authentication domain **410-b** in the authentication hierarchy **400**, and the DMS **110** may refrain from outputting the audit information accordingly.

(76) The multi-tenancy DMS **110** described herein may thereby utilize an authentication hierarchy **400** (e.g., a table of information, pointers, or other authorization information) to determine whether a user **420** of the DMS **110** is authorized to access audit information and other data associated with other users **420** of the DMS **110**. Techniques for authorizing access to audit information are described in further detail elsewhere herein, including with reference to FIGS. **4-6**.

(77) FIG. **5** illustrates an example of a computing environment **500** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The computing environment **500** may implement or be implemented by aspects of the computing environment **100**, the multi-tenancy system **200**, and the authentication hierarchy **400** described with reference to FIGS. **1-3**. For example, the computing environment **500** includes a DMS **510** and a computing device **515**, which may be in communication with one another via a network **520**, as described with reference to FIG. **1**. The DMS **510** may provide backup services for multiple tenants (e.g., organizations or business units). The tenants of the DMS **510** may be organized in a hierarchical structure, as described with reference to FIG. **2**. In this example, the DMS **510** may determine whether a first user has permission to access authorization information for another entity (e.g., a user or computing object **540**) based on a tenant associated with a log-in session by the first user, while accounting for impacts of a multi-level hierarchical multi-tenancy environment.

(78) The DMS **510** may provide backup services for data sources of multiple tenants. The data sources that are to be backed up may be stored within one or more computing systems **105** (not pictured in FIG. **5**). The DMS **510** may include or be coupled with a central manager **525**. The central manager **525** may be a component of the DMS **510** that manages or facilitates tasks performed by the DMS **510**. In this example, the central manager **525** may manage access authorization and filtering for authorization information in a multi-tenancy system (e.g., an audit authorization component).

(79) The DMS **510** may include multiple computing objects **540** (e.g., N computing objects **540**), which may be resources or other entities within the DMS **510**. In some examples, the DMS **510** may include one or more clusters (e.g., storage clusters or storage nodes), and the computing objects **540** may represent examples of a cluster or a portion of a cluster of the DMS **510**. The computing objects may be, for example, virtual machines, folders, filesets, databases, other computing objects, or any combination thereof. Each computing object **540** of the DMS **510** may

be associated with or identified by a respective object ID (e.g., a unique ID). The computing objects **540** may be stored at the DMS **510**, at the central manager **525**, at one or more other servers or databases associated with or coupled to the DMS **510**, or any combination thereof. The computing objects **540** may be organized into a computing object hierarchy, as described in further detail elsewhere herein, including with reference to FIG. 3.

(80) The DMS **510** may obtain audit information for users of the DMS **510**. Each user may log in to the DMS **510** (e.g., via a user interface using a computing device **515**). Each log-in session may be associated with a respective tenant. For example, a user (e.g., an administrator or some other user of a tenant) may log in to the DMS **510** as part of or using the respective tenant. In some examples, a single user may be able to establish multiple log-in sessions with the DMS **510**, each log-in session associated with a different tenant. The DMS **510** may monitor events associated with and operations performed by each user that is logged in to the DMS **510**. The DMS **510** may maintain a log or record of the events and operations performed by each user when logged in to the DMS **510**. The log may be stored in the form of audit information.

(81) The DMS **510** may generate and store audit information for a given user in response to detecting an audit event. An audit event may include an event or operation performed by the user as part of a log-in session with the DMS **510**. An audit event may include, for example, a user logging in to the DMS **510** and initiating a log-in session, a user deleting a file or some other resources when logged in to the DMS **510**, a user modifying one or more configurations or settings (e.g., service level agreement configurations), other critical actions performed by the user when logged in to the DMS **510**, or any combination thereof. The DMS **510** may determine which events and operations include auditable events. In some examples, the events that count as auditable events may be different for different tenants. The DMS **510** may monitor operations performed by each user to determine whether an audit event has been performed.

(82) In response to detecting an audit event performed by a user of the DMS **510**, the DMS **510** may generate audit information to store a record of that audit event occurring. The audit information may include, for example, data or metadata that indicates a time of the audit event, a type of the audit event, a duration of the audit event, a user ID of the user that performed the audit event, an ID of an entity that was the object or target of the audit event, or any combination thereof. The DMS **510** may store the audit information in an audit information database **530**. In some examples, the audit information may indicate a subject and an object of the audit event. The subject may represent a user or other entity that performs the audit event, and the object may represent a user, computing object, or other entity that is the target of the event (e.g., on which the audit operation is performed). For example, if a first user updates one or more parameters or settings associated with a computing object **540**, the DMS **510** may generate and store audit information that records information about the audit event and that the first user is the subject and the computing object **540** is the object of the audit event.

(83) As described herein, when the DMS **510** generates audit information for a given user, the DMS **510** may analyze context information for a log-in session of the user. The context information may indicate a user ID of the user, a tenant ID of a tenant to which the user is logged in, other information, or any combination thereof. The DMS **510** may store the tenant ID associated with the user with the audit information in the audit information database **530**. For example, each section or portion of audit information (e.g., audit information generated during each audit) may be indexed by a respective tenant ID associated with a tenant to which the user being audited was logged in. By storing the tenant ID with the audit information in the audit information database **530**, the DMS **510** may efficiently and reliably determine which tenant is associated with the audit information, which may improve security and reduce latency associated with backup and management of data by the DMS **510**.

(84) In some examples, the DMS **510** may store the audit information in a table format in the audit information database **530**. For example, the DMS **510** may store the audit information in one or

more rows, where each row may include audit information for a respective audit and a corresponding tenant ID. In some examples, multiple rows of audit information may include a same tenant ID if, for example, the DMS **510** audits multiple users of a same tenant, or audits a single user multiple times (e.g., if a user performs multiple audit events). In the example illustrated in FIG. 5, a first row of the audit information database **530**) may include audit information for an audit of a user logged in to or associated with a first tenant having a tenant ID #1. A second row and a third row of the audit information database **530** may include audit information for an audit of one or two users logged in to or associated with a second tenant having a tenant ID #2. Other rows of audit information may include audit information for audits of users logged in to the first tenant, the second tenant, or any other tenant of the DMS **510**. In some examples, the DMS **510** may additionally or alternatively store the ID of the user, the ID of the entity that is the object of the audit event, or both with the audit information.

(85) In some examples, a first user of the DMS **510** may request to view audit information for another entity of the DMS **510**, such as a second user or a computing object. The first user may be logged in to the DMS **510** via a tenant of the multiple tenants supported by the DMS **510**. For example, the first user may be an administrator or other user of a tenant. As part of the log-in session with the DMS **510**, the first user may output (e.g., send, transmit) an audit information request **545** to the DMS **510**. The first user may utilize a computing device **515** to log in to the DMS **510** and access a user interface that displays account settings and information associated with the first user's log-in session. The first user may send the audit information request **545** via the user interface and the network **520** to the DMS **510**. The audit information request **545** may indicate a request for the first user to view or access audit information and may indicate an entity associated with the requested audit information. For example, the first user may request to view auditable events that were performed by a second user when logged in to the DMS **510**, or auditable events that were performed on the second user, or auditable events that were performed on a set of one or more computing objects **540**, or any combination thereof.

(86) Techniques, systems, and devices described herein provide for the DMS **510** to determine whether the first user that sent the audit information request **545** has permission to access the requested audit information for the entity based on a first tenant associated with the log-in session by the first user, while accounting for impacts of a multi-level hierarchical multi-tenancy environment. In response to receiving the audit information request **545**, the DMS **510** may determine an ID of the first user and an ID of the first tenant to which the first user is logged in based on context information for the log-in session via which the audit information request **545** is sent. The context information may include information that defines or identifies the log-in session of the first user, including the user ID, the tenant ID, other context information, or any combination thereof.

(87) After determining the tenant ID associated with the audit information request **545**, the DMS **510** (e.g., the central manager **525**) may identify authorization information associated with the tenant. The authorization information may indicate that the tenant has access to an authentication domain or other node or entity within an authentication hierarchy associated with multiple tenants supported by the DMS **510**. For example, the DMS **510** may scan an authentication hierarchy associated with the tenants supported by the DMS **510** and determine a node in the authentication hierarchy to which the tenant has access. The node may be, for example, an authentication domain, a user, or a user group of one or more users. The tenant may have authorization over or permission to access all nodes (e.g., users, user groups, subtenants) below the node or authentication domain in the authentication hierarchy. The authentication hierarchy may represent an example of the authentication hierarchy **400** described with reference to FIG. 4. Additionally, or alternatively, the authorization information may indicate a set of one or more computing objects **540** to which the tenant has access within a computing object hierarchy, such as the computing object hierarchy **300** described with reference to FIG. 3.

(88) If the first user requests to view audit information associated with a second user, the DMS 510 may determine a hierarchical relationship between the second user indicated via the audit information request 545 and the authentication domain to which the tenant associated with the audit information request 545 has access in the authentication domain. That is, the DMS 510 may determine whether the second user is included in or below the authentication domain in the authentication hierarchy. The DMS 510 may determine whether the first user is authorized or permitted to access the requested audit information for the second user based on the hierarchical relationship.

(89) If the second user is below the authentication domain in the authentication hierarchy, the DMS 510 may determine that the tenant associated with the audit information request 545 is authorized to access the second user, and the DMS 510 may output (e.g., send, transmit, export, mount) the requested audit information for the second user to the first user via the network 520 and the computing device 515. The first user may determine whether to adjust one or more privacy settings or configurations of the second user based on the audit information. For example, if the first user is an administrator of the tenant, and the second user is a user of the same tenant or a subtenant of the tenant, the audit information may indicate whether the second user is in compliance with security guidelines and settings for the tenant. If the audit information indicates that the second user performed an unauthorized operation, the first user may remove the second user from the tenancy chain. That is, the first user may request that the DMS 510 revoke access by the second user to the subtenant, the tenant, or both, and any computing resources and other data associated with the subtenant and the tenant.

(90) If the second user is not below or included in the authentication domain in the authentication hierarchy, the DMS 510 may determine that the tenant associated with the audit information request 545 is not authorized to access the second user, and the DMS 510 may refrain from outputting the requested audit information for the second user to the first user.

(91) In some examples, if the audit information request 545 requests audit information associated with one or more computing objects 540, the DMS 510 may determine a hierarchical relationship between the one or more computing objects 540 indicated via the audit information request 545 and a set of one or more computing objects 540 to which the tenant has access in a computing object hierarchy. If the requested computing objects 540 are included in or below the set of one or more computing objects 540 to which the tenant has access in a computing object hierarchy, the DMS 510 may output (e.g., send, transmit, export, mount) the requested audit information for the one or more computing objects 540 indicated via the audit information request 545 to the first user via the network 520 and the computing device 515. The first user may adjust one or more settings or configurations associated with the computing objects 540 based on an analysis of the audit information.

(92) If the requested computing objects 540 are not included in or below the set of one or more computing objects 540 to which the tenant has access in a computing object hierarchy, the DMS 510 may determine that the first user is not authorized to access the requested audit information, and the DMS 510 may refrain from outputting the requested audit information for the one or more computing objects 540 indicated via the audit information request 545 to the first user.

(93) The DMS 510 may similarly verify authorization for any audit information request 545 received from any other user of the DMS 510. The DMS 510 may thereby ensure audit information for users of a multi-tenancy system is secure and reliable by verifying authorization of a user to access the audit information before outputting the audit information to the user, which may improve reliability of the DMS 510

(94) In some examples, a user of the DMS 510 may request to filter audit information by one or more parameters, such a type of audit information, per tenant, per subject, per object, or some other filtering criteria. As described herein, the DMS 510 may use the tenant IDs stored in the audit information database 530 to support filtering the audit information in the audit information

database **530** per tenant. In some examples, a user of a relatively high level tenant (e.g., an administrator of a global organization, MSP, or other tenant that has authorization over at least one subtenant) may request to view audit information associated with users of one or more subtenants. After establishing or initiating a log-in session with the DMS **510** via the tenant, the user may access a user interface via the computing device **515**. The user may perform a search or other request in an audit page of the user interface to view audit information for one or more subtenants. The DMS **510** may receive the filtering request **550** via the network **520**. The filtering request **550** may indicate an ID of the subtenant(s) for which the user requests to view audit information.

(95) In response to receiving the filtering request **550** to view audit information for a given subtenant, the DMS **510** (e.g., the central manager **525** or some other component of the DMS **510**) may filter the audit information in the audit information database **530** per tenant. For example, the DMS **510** may filter or select audit information that is associated with users of the subtenant indicated via the filtering request **550** from among all of the audit information stored in the audit information database **530**. To perform the filtering, the DMS **510** may scan the audit information database **530** and identify each row that includes a tenant ID of the subtenant indicated via the filtering request **550**. The DMS **510** may select the audit information that includes the tenant ID of the subtenant and may output that audit information to the first user via the user interface (e.g., via the network **520** and the computing device **515**).

(96) In the example illustrated in FIG. 5, if the filtering request **550** requests to view audit information for a tenant having the tenant ID #1, the DMS **510** may select the first and final rows of audit information in the audit information database **530** (e.g., and any other rows not pictured in FIG. 5 that include the tenant ID #1). The DMS **510** may output the selected audit information to the first user in response to the filtering request **550**. In some examples, the DMS **510** may display the audit information in a list or other organized format. The DMS **510** may be operable to paginate, sort, or otherwise further filter the audit information in the user interface based on one or more other criteria. By storing tenant IDs with audit information, the DMS **510** may thereby provide filtered audit information to a user of the DMS **510**.

(97) In some examples, the DMS **510** may verify authorization of the user to access the audit information before outputting the filtered audit information. For example, the DMS **510** may use the tenant ID of the tenant associated with the first user to determine whether the second user is below an authentication domain that the tenant is authorized to access in the authentication hierarchy. If the second user is not below the authentication domain, the DMS **510** may refrain from outputting the filtered audit information to the first user. second user is not below the authentication domain, the DMS **510** may output the filtered audit information to the first user.

(98) The DMS **510** described herein may thereby obtain and store audit information for multiple users of the DMS **510** while managing client data for multiple tenants. By using the described techniques for authorizing access to audit information and filtering audit information, the DMS **510** may support efficient, secure, and reliable management of audit information in a multi-tenant system.

(99) FIG. 6 illustrates an example of a process flow **600** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The process flow **600** may implement or be implemented by aspects of the computing environments **100** and **500**, the multi-tenancy system **200**, the computing object hierarchy **300**, and the authentication hierarchy **400**, as described with reference to FIGS. 1-5. For example, the process flow **600** may be implemented by a first user **605**, which may be logged in to a tenant **615** (e.g., an administrator for the tenant **615**, or some other user **605** of the tenant **615**) and one or more components of a DMS **610**, which may represent examples of corresponding devices and components as described with reference to FIGS. 1-5. In some examples, the first user **605** may represent an example of a computing device that may communicate with the DMS **610** via a user interface, such as the computing devices **115** or **515** described with reference to FIGS. 1 and 5. In

this example, the DMS **610** may determine whether the first user **605** has permission to access audit information for a second user of the multi-tenancy DMS **610** while accounting for impacts of a multi-level hierarchical multi-tenancy environment.

(100) In some aspects, the operations illustrated in the process flow **600** may be performed by hardware (e.g., including circuitry, processing blocks, logic components, and other components), code (e.g., software or firmware) executed by a processor, or any combination thereof. For example, aspects of the process flow **600** may be implemented or managed by a DMS, an audit authorization component, or some other software or application within a DMS **610** that is configured to manage backup and restoration of data and other computing resources for multiple tenants.

(101) Although the first user **605** and the DMS **610** are illustrated in FIG. **6**, it is to be understood that the process to authorize access for audit information in a multi-tenancy system as described herein may be performed by any one or more components of a DMS **610**, including components not shown in FIG. **6**. For example, the DMS **610** may include or be coupled with a central manager, one or more other components, or any combination thereof (not pictured in FIG. **6**) that may be configured to facilitate or manage any one or more of the processes illustrated in FIG. **6**.

Additionally, or alternatively, one or more components may perform tasks or may be configured to function in a different manner than illustrated in FIG. **6**. Alternative examples of the following may be implemented, where some steps are performed in a different order than described or are not performed at all. In some cases, steps may include additional features not mentioned below, further steps may be added, or different components may perform the steps.

(102) At **620**, in some examples, the DMS **610** may identify an audit event for an entity of the DMS **610**. The DMS may monitor operations performed by the users of the DMS **610** when the users are logged in to the DMS **610**. Identifying the audit event may include identifying a log-in operation, an operation to change one or more service level agreement settings, audit tasks performed by a user, a delete operation, an operation to change a configuration of a user, one or more other auditable operations, or any combination thereof. The entity may be a subject of the audit event, such as the first user **605** or some other user of the DMS **610** that performs the audit event. Additionally, or alternatively, the entity may be an object (e.g., target) of the audit event, such as a user or an object that is being modified or changed by the audit event. In some examples, the DMS **610** may dynamically configure what types of operations count as auditable events. After identifying the audit event, the DMS **610** may identify context information for a log-in session associated with the user being audited (e.g., the subject of the audit event). The context information may include a user ID of the user and a tenant ID of a tenant to which the user is logged in.

(103) At **625**, in some examples, after identifying the audit event and the corresponding context information, the DMS **610** may store audit information that indicates the audit event. The DMS **610** may store the tenant ID with the audit information, in some examples. The DMS **610** may thereby store a tenant ID mapped to each portion of audit information in an audit information database of the DMS **610**, such as the audit information database **530**) described with reference to FIG. **5**.

(104) At **630**, the first user **605** may output (e.g., transmit, send) a request, to the DMS **610** via a user interface, to provide the first user **605** with audit information associated with an entity of the DMS **610**. In some examples, the first user **605** may be an administrator or other user of the tenant **615**, and the tenant **615** may be a higher-level tenant in a hierarchy of multiple tenants of the DMS **610**. The DMS **610** may include multiple computing objects and may provide protection for data sources associated with the multiple tenants. The entity may be a subject, such as a second user or an object, such as a second user or a computing object that is a target of the audit information.

(105) At **635**, the DMS **610** may identify, in response to the request, context information for a log-in session associated with the request. For example, the first user **605** may log in to the DMS **610** to initiate the log-in session. The first user **605** may transmit the request as part of the log-in session. The first user **605** may log in to the DMS **610** as a user of the tenant **615** (e.g., an administrator or

other user of the tenant **615**). The context information for the log-in session may include a tenant ID of the tenant **615** associated with the request. The Tenant ID may identify or distinguish the tenant **615** from among the other tenants of the multi-tenancy DMS **610**.

(106) At **640**, the DMS **610** may identify, based on the context information, authorization information associated with the tenant. In some examples, if the DMS **610** is performing access control based on a second user associated with the audit information (e.g., a subject or an object of the audit event), the DMS **610** may scan an authentication hierarchy associated with the multiple tenants of the DMS **610** using the tenant ID of the tenant **615** associated with the request to identify authorization information. The authentication hierarchy may represent a hierarchy of authentication domains between tenants, user groups, and users, as described in further detail elsewhere herein, including with reference to FIG. 4. In some examples, scanning the authentication hierarchy may include scanning a database or table of information that represents the hierarchy (e.g., a “flattened” hierarchy). The authorization information associated with the tenant **615** may indicate that the tenant **615** has access to one or more nodes within the authentication hierarchy. The one or more nodes may be an authentication domain, a user of the DMS **610**, or a user group of the DMS **610**, as described with reference to FIG. 4.

(107) In some other examples, if the DMS **610** is performing access control based on a computing object associated with the audit information (e.g., an object of the audit event), the DMS **610** may scan a computing object hierarchy associated with the DMS **610**, such as the computing object hierarchy **300** described with reference to FIG. 3, to identify the authorization information. The authorization information may include a set of one or more computing objects to which the tenant **615** associated with the request has access in the computing object hierarchy.

(108) At **645**, the DMS **610** may determine whether to output, to the first user **605** in response to the request, the audit information associated with the entity based on a hierarchical relationship, within a hierarchy of the DMS **610** (e.g., the authentication hierarchy or the computing object hierarchy) between the entity and the tenant **615** associated with the request. For example, if the first user **605** requests audit information associated with a second user, and the DMS **610** performs access control based on the subject or object of audit information, the DMS **610** may determine whether the tenant **615** has access to the second user or an authentication domain that includes the second user in the authentication hierarchy. If the first user **605** requests audit information associated with at least one computing object, and the DMS **610** performs access control based on the object of audit information, the DMS **610** may determine whether the at least one computing object is included in a set of one or more computing objects to which the tenant **615** has access in a computing object hierarchy. In some examples, the DMS **610** may perform access control based on one or both of the subject and the object of the audit information.

(109) At **650**, in some examples, the DMS **610** may output (e.g., send, transmit), to the first user **605** via the user interface and in response to the request, the audit information associated with the entity. The DMS **610** may output the audit information to the first user **605** if the hierarchical relationship indicates that the second user is below the authentication domain within the authentication hierarchy or that the at least one computing object is included in the set of one or more computing objects to which the tenant **615** has access in the computing object hierarchy. That is, the DMS **610** may output the audit information to the first user **605** (e.g., display the audit information via a user interface) if the tenant **615** to which the first user **605** is logged in has authorization to access the entity, such as the second user or the at least one computing object, or both.

(110) In some examples, the DMS **610** may refrain from outputting, to the first user **605** in response to the request, the audit information associated with the entity. The DMS **610** may refrain from outputting the audit information if the hierarchical relationship indicates that the entity (e.g., a second user) is not below the authentication domain within the authentication hierarchy, or if the entity (e.g., at least one computing object) is not assigned to the tenant in a computing object

hierarchy, or both. That is, the DMS **610** may refrain from outputting the audit information to the first user **605** if the tenant **615** to which the first user **605** is logged in does not have authorization to access the second user, or the at least one computing object, or both.

(111) The DMS **610** may thereby facilitate secure and reliable exchanges of audit information with customers or users **605** of the DMS **610**. The DMS **610** may use context information associated with a log-in session of a user **605** to determine a tenant **615** to which the user **605** is logged in. The DMS **610** may use a tenant ID of the tenant **615** to determine whether the tenant **615** is authorized or permitted to access the entity associated with requested audit information. The DMS **610** may thereby support multiple tenants while maintaining reliable and secure data management services for each tenant and its corresponding users separately.

(112) FIG. 7 illustrates an example of a process flow **700** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The process flow **700** may implement or be implemented by aspects of the computing environments **100** and **500**, the multi-tenancy system **200**, the computing object hierarchy **300**, the authentication hierarchy **400**, and the process flow **700**, as described with reference to FIGS. 1-6. For example, the process flow **700** may be implemented by a user **705**, which may be logged in to a first tenant **715** (e.g., an administrator for the first tenant **715**, or some other user **705** of the first tenant **715**) and one or more components of a DMS **710**, which may represent examples of corresponding devices and components as described with reference to FIGS. 1-5. In some examples, the user **705** may represent an example of a computing device that may communicate with the DMS **710** via a user interface, such as the computing devices **115** or **515** described with reference to FIGS. 1 and 5. In this example, the DMS **710** may generate and store audit information associated with multiple users of the multi-tenancy DMS **710**, and the user **705** may request the DMS **710** to filter audit information per tenant.

(113) In some aspects, the operations illustrated in the process flow **700** may be performed by hardware (e.g., including circuitry, processing blocks, logic components, and other components), code (e.g., software or firmware) executed by a processor, or any combination thereof. For example, aspects of the process flow **700** may be implemented or managed by a DMS **710**, an audit authorization component, or some other software or application within a DMS **710** that is configured to manage backup and restoration of data and other computing resources for multiple tenants.

(114) Although the user **705** and the DMS **710** are illustrated in FIG. 7, it is to be understood that the process to authorize access for audit information in a multi-tenancy system as described herein may be performed by any one or more components of a DMS **710**, including components not shown in FIG. 7. For example, the DMS **710** may include or be coupled with a central manager, one or more other components, or any combination thereof (not pictured in FIG. 7) that may be configured to facilitate or manage any one or more of the processes illustrated in FIG. 7.

Additionally, or alternatively, one or more components may perform tasks or may be configured to function in a different manner than illustrated in FIG. 7. Alternative examples of the following may be implemented, where some steps are performed in a different order than described or are not performed at all. In some cases, steps may include additional features not mentioned below, further steps may be added, or different components may perform the steps.

(115) At **720**, the user **705** may output (e.g., transmit, send) a request to view audit information associated with one or more users of a second tenant of the DMS **710** (not pictured in FIG. 7). For example, the user **705** may be an administrator or some other user **705** of the first tenant **715**, and the first tenant **715** may be a higher level tenant within a hierarchy of the multiple tenants supported by the DMS **710** (e.g., a global organization, an MSP, or some other higher level tenant) than the second tenant. The second tenant may be a subtenant of the first tenant **715**. In some examples, the user **705** may request to view all audit information previously obtained by and stored by the DMS **710** that is associated with any user of the second tenant, or the user **705** may request

to view all audit information previously obtained by and stored by the DMS **710** that is associated with all users of any subtenant of the first tenant **715** that are assigned to or below the first tenant **715** in the tenant hierarchy (e.g., the second tenant and any other subtenants of the first tenant **715**). (116) In some examples, the user **705** may access a user interface of the DMS **710** and perform a search within the user interface. If the user **705** searches for a given organization or subtenant, the search may correspond to or trigger the request for audit information associated with the given organization or subtenant. For example, the user **705** may request to view a display on the user interface of the audit information for the given subtenant.

(117) At **725**, the DMS **710** may filter a database of audit information stored by the DMS **710**. The DMS **710** may filter the audit information per tenant or group of tenants and in response to the request from the user **705**, one or more other requests from other users of the DMS **710**, or any combination thereof.

(118) As part of filtering the database of audit information for the second tenant in response to the request, at **730**, the DMS **710** may scan the database of audit information and select audit information from within the database that includes a tenant ID of the second tenant. That is, the DMS **710** may select audit information associated with users that were logged in to the second tenant at the time the audit was performed. The DMS **710** may refrain from selecting, or may filter out, other audit information in the audit information database that is associated with other tenants different than the second tenant indicated via the request. By storing audit information with corresponding tenant IDs, the DMS **710** may thereby filter the audit information database per tenant, which may improve reliability and efficiency of the services provided by the DMS **710**.

(119) At **735**, the DMS **710** may output (e.g., transmit, send) the audit information associated with the second tenant based on filtering the database of audit information. The DMS **710** may output the audit information that was selected by the DMS **710** and includes the tenant ID of the second tenant. In some examples, the DMS **710** may display the audit information for the second tenant as a list or set via a user interface. For example, the DMS **710** (e.g., a backend of the audit query) may support sorting, pagination, and filtering of the audit information. In some examples, before outputting the audit information, the DMS **710** may determine whether the user **705** is authorized to access the requested audit information based on a hierarchical relationship between the first tenant **715** to which the user **705** is logged in and the second user, as described in further detail elsewhere herein, including with reference to FIG. **6**.

(120) In some examples, although not illustrated in FIG. **7**, the user **705** may request to view audit information associated with two or more other tenants or subtenants. The DMS **710** may similarly filter the audit information database per group of tenants and output the audit information for the multiple tenants to the user **705**. The DMS **710** may additionally, or alternatively, support filtering and pagination of the audit information database per subject of the audit information, per object of the audit information, per user, or per any other audit information parameter or entity.

(121) By storing tenant IDs with corresponding audit information, the DMS **710** may thereby support filtering audit information per tenant as described herein. The described techniques may improve security, reliability, and customer satisfaction while accounting for impacts of a multi-level hierarchical multi-tenancy environment.

(122) FIG. **8** illustrates a block diagram **800** of a system **805** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The system **805** may be an example of aspects of a DMS as described herein. The system **805** may include an input interface **810**, an output interface **815**, and an audit authorization component **820**. The system **805** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(123) The input interface **810** may manage input signaling for the system **805**. For example, the input interface **810** may receive input signaling (e.g., messages, packets, data, instructions, commands, or any other form of encoded information) from other systems or devices. The input

interface **810** may send signaling corresponding to (e.g., representative of or otherwise based on) such input signaling to other components of the system **805** for processing. For example, the input interface **810** may transmit such corresponding signaling to the audit authorization component **820** to support access authorization for audit information in a multi-tenancy DMS. In some cases, the input interface **810** may be a component of a network interface **1125** as described with reference to FIG. **11**.

(124) The output interface **815** may manage output signaling for the system **805**. For example, the output interface **815** may receive signaling from other components of the system **805**, such as the audit authorization component **820**, and may transmit such output signaling corresponding to (e.g., representative of or otherwise based on) such signaling to other systems or devices. In some cases, the output interface **815** may be a component of a network interface **1125** as described with reference to FIG. **11**.

(125) The audit authorization component **820**, the input interface **810**, the output interface **815**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of access authorization for audit information in a multi-tenancy DMS as described herein. For example, the audit authorization component **820**, the input interface **810**, the output interface **815**, or various combinations or components thereof may support a method for performing one or more of the functions described herein.

(126) In some examples, the audit authorization component **820**, the input interface **810**, the output interface **815**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include a processor, a DSP, an ASIC, an FPGA or other programmable logic device, a discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting a means for performing the functions described in the present disclosure. In some examples, a processor and memory coupled with the processor may be configured to perform one or more of the functions described herein (e.g., by executing, by the processor, instructions stored in the memory).

(127) Additionally, or alternatively, in some examples, the audit authorization component **820**, the input interface **810**, the output interface **815**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by a processor. If implemented in code executed by a processor, the functions of the audit authorization component **820**, the input interface **810**, the output interface **815**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting a means for performing the functions described in the present disclosure).

(128) In some examples, the audit authorization component **820** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the input interface **810**, the output interface **815**, or both. For example, the audit authorization component **820** may receive information from the input interface **810**, send information to the output interface **815**, or be integrated in combination with the input interface **810**, the output interface **815**, or both to receive information, transmit information, or perform various other operations as described herein.

(129) For example, the audit authorization component **820** may be configured as or otherwise support a means for receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The audit authorization component **820** may be configured as or otherwise support a means for identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The audit authorization component **820** may be configured as or otherwise

support a means for identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The audit authorization component **820** may be configured as or otherwise support a means for determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(130) By including or configuring the audit authorization component **820** in accordance with examples as described herein, the system **805** (e.g., a processor controlling or otherwise coupled with the input interface **810**, the output interface **815**, the audit authorization component **820**, or a combination thereof) may support techniques for improved data management reliability, improved DMS performance and compliance for applications and systems that use objects managed by the system **805**, reduced power consumption, reduced processing complexity.

(131) FIG. **9** illustrates a block diagram **900** of a system **905** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. In some examples, the system **905** may be an example of aspects of one or more components described with reference to FIG. **1**, such as a DMS **110**. The system **905** may be an example of aspects of a system **805** or a DMS **110** as described herein. The system **905** may include an input interface **910**, an output interface **915**, and an audit authorization component **920**. The system **905** may also include one or more processors. Each of these components may be in communication with one another (e.g., via one or more buses, communications links, communications interfaces, or any combination thereof).

(132) The input interface **910** may manage input signaling for the system **905**. For example, the input interface **910** may receive input signaling (e.g., messages, packets, data, instructions, commands, or any other form of encoded information) from other systems or devices. The input interface **910** may send signaling corresponding to (e.g., representative of or otherwise based on) such input signaling to other components of the system **905** for processing. For example, the input interface **910** may transmit such corresponding signaling to the audit authorization component **920** to support access authorization for audit information in a multi-tenancy DMS. In some cases, the input interface **910** may be a component of a network interface **1125** as described with reference to FIG. **11**.

(133) The output interface **915** may manage output signaling for the system **905**. For example, the output interface **915** may receive signaling from other components of the system **905**, such as the audit authorization component **920**, and may transmit such output signaling corresponding to (e.g., representative of or otherwise based on) such signaling to other systems or devices. In some cases, the output interface **915** may be a component of a network interface **1125** as described with reference to FIG. **11**.

(134) The system **905**, or various components thereof, may be an example of means for performing various aspects of access authorization for audit information in a multi-tenancy DMS as described herein. For example, the audit authorization component **920** may include a request component **925**, a context information component **930**, an authorization information component **935**, an access authorization component **940**, or any combination thereof. The audit authorization component **920** may be an example of aspects of an audit authorization component **820** as described herein. In some examples, the audit authorization component **920**, or various components thereof, may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the input interface **910**, the output interface **915**, or both. For example, the audit authorization component **920** may receive information from the input interface **910**, send information to the output interface **915**, or be integrated in combination with the input interface **910**, the output interface **915**, or both to receive information, transmit information, or perform various other operations as described herein.

(135) The request component **925** may be configured as or otherwise support a means for receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The context information component **930** may be configured as or otherwise support a means for identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The authorization information component **935** may be configured as or otherwise support a means for identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The access authorization component **940** may be configured as or otherwise support a means for determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(136) FIG. **10** illustrates a block diagram **1000** of an audit authorization component **1020** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The audit authorization component **1020** may be an example of aspects of an audit authorization component **820**, an audit authorization component **920**, or both, as described herein. The audit authorization component **1020**, or various components thereof, may be an example of means for performing various aspects of access authorization for audit information in a multi-tenancy DMS as described herein. For example, the audit authorization component **1020** may include a request component **1025**, a context information component **1030**, an authorization information component **1035**, an access authorization component **1040**, an audit information component **1045**, an audit event component **1050**, an audit information storage component **1055**, an audit information filtering component **1060**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses, communications links, communications interfaces, or any combination thereof).

(137) The request component **1025** may be configured as or otherwise support a means for receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The context information component **1030** may be configured as or otherwise support a means for identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The authorization information component **1035** may be configured as or otherwise support a means for identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The access authorization component **1040** may be configured as or otherwise support a means for determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(138) In some examples, the audit information component **1045** may be configured as or otherwise support a means for obtaining the audit information associated with the entity, where entries within the audit information are associated with respective events including respective actions performed by the entity on respective objects, and where determining whether to output the audit information associated with the entity is further based on whether the tenant associated with the request has

access to audit information associated with the respective objects.

(139) In some examples, the audit information component **1045** may be configured as or otherwise support a means for obtaining the audit information associated with the entity, where entries within the audit information are associated with respective events including respective actions performed by respective subjects on the entity, and where determining whether to output the audit information associated with the entity is further based on whether the tenant associated with the request has access to audit information associated with the respective subjects.

(140) In some examples, the authorization information indicates that the tenant has access to an authentication domain within the hierarchy, the hierarchy including an authentication hierarchy associated with the set of multiple tenants. In some examples, the hierarchical relationship indicates a relationship between the entity and the authentication domain in the authentication hierarchy.

(141) In some examples, the audit information component **1045** may be configured as or otherwise support a means for outputting, to the first user in response to the request, the audit information associated with the entity based on the hierarchical relationship indicating that the entity is below the authentication domain within the authentication hierarchy.

(142) In some examples, the audit information component **1045** may be configured as or otherwise support a means for refraining from outputting, to the first user in response to the request, the audit information associated with the entity based on the hierarchical relationship indicating that the entity is not below the authentication domain within the authentication hierarchy.

(143) In some examples, the authentication domain includes a node within the authentication hierarchy, the node above a single user of a set of multiple users of the DMS in the authentication hierarchy. In some examples, the authentication domain includes a node within the authentication hierarchy, the node above a user group of the DMS in the authentication hierarchy.

(144) In some examples, the authorization information indicates a set of one or more objects to which the tenant associated with the request has access within the hierarchy, the hierarchy including an object hierarchy associated with a set of multiple objects of the DMS. In some examples, the hierarchical relationship indicates a relationship between the entity and the set of one or more objects in the object hierarchy

(145) In some examples, the audit event component **1050** may be configured as or otherwise support a means for identifying an audit event for a third user of the DMS. In some examples, the context information component **1030** may be configured as or otherwise support a means for identifying, in response to identifying the audit event, second context information for a second log-in session associated with the third user, where the second context information for the second log-in session includes a second ID of a second tenant associated with the third user, the second tenant included in the set of multiple tenants. In some examples, the audit information storage component **1055** may be configured as or otherwise support a means for storing, in a database, second audit information associated with the audit event and the second ID of the second tenant associated with the third user.

(146) In some examples, the audit event component **1050** may be configured as or otherwise support a means for monitoring one or more operations performed by the third user of the DMS based on the third user initiating the second log-in session, where identifying the audit event includes identifying an operation of the one or more operations.

(147) In some examples, the request component **1025** may be configured as or otherwise support a means for receiving, from the first user as part of the log-in session associated with the tenant, a second request to view second audit information associated with one or more users of one or more other, where the one or more other tenants include one or more subtenants of the tenant within a tenant hierarchy associated with the set of multiple tenants. In some examples, the audit information filtering component **1060** may be configured as or otherwise support a means for filtering, per tenant or group of tenants and in response to the second request, a database of audit information stored by the DMS. In some examples, the audit information filtering component **1060**

may be configured as or otherwise support a means for outputting, to the first user via a user interface in response to the second request, the second audit information associated with the one or more users of the one or more other tenants based on the filtering and the one or more other tenants including the one or more subtenants of the tenant within the tenant hierarchy.

(148) In some examples, to support filtering the database of audit information per tenant or group of tenants, the audit information filtering component **1060** may be configured as or otherwise support a means for selecting the second audit information from among the database of audit information based on the second audit information including the ID of the tenant associated with the request, where remaining audit information in the database includes IDs of other tenants included in the set of multiple tenants.

(149) In some examples, the first user initiates a second log-in session with the DMS after the log-in session, and second context information for the second log-in session associated with the first user includes a second ID of a second tenant included in the set of multiple tenants, the second tenant different than the tenant associated with the request.

(150) FIG. **11** illustrates a block diagram **1100** of a system **1105** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The system **1105** may be an example of or include the components of a system **805**, a system **905**, or a DMS as described herein. The system **1105** may include components for data management, including components such as an audit authorization component **1120**, an input information **1110**, an output information **1115**, a network interface **1125**, a memory **1130**, a processor **1135**, and a storage **1140**. These components may be in electronic communication or otherwise coupled with each other (e.g., operatively, communicatively, functionally, electronically, electrically: via one or more buses, communications links, communications interfaces, or any combination thereof). Additionally, the components of the system **1105** may include corresponding physical components or may be implemented as corresponding virtual components (e.g., components of one or more virtual machines). In some examples, the system **1105** may be an example of aspects of one or more components described with reference to FIG. **1**, such as a DMS **110**.

(151) The network interface **1125** may enable the system **1105** to exchange information (e.g., input information **1110**, output information **1115**, or both) with other systems or devices (not shown). For example, the network interface **1125** may enable the system **1105** to connect to a network (e.g., a network **120** as described herein). The network interface **1125** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. In some examples, the network interface **1125** may be an example of aspects of one or more components described with reference to FIG. **1**, such as one or more network interfaces **165**.

(152) Memory **1130** may include RAM, ROM, or both. The memory **1130** may store computer-readable, computer-executable software including instructions that, when executed, cause the processor **1135** to perform various functions described herein. In some cases, the memory **1130** may contain, among other things, a basic input/output system (BIOS), which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some cases, the memory **1130** may be an example of aspects of one or more components described with reference to FIG. **1**, such as one or more memories **175**.

(153) The processor **1135** may include an intelligent hardware device, (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, a field programmable gate array (FPGA), a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). The processor **1135** may be configured to execute computer-readable instructions stored in a memory **1130** to perform various functions (e.g., functions or tasks supporting access authorization for audit information in a multi-tenancy DMS). Though a single processor **1135** is depicted in the example of FIG. **11**, it is to be understood that the system **1105** may include any quantity of one or more of processors **1135** and that a group of

processors **1135** may collectively perform one or more functions ascribed herein to a processor, such as the processor **1135**. In some cases, the processor **1135** may be an example of aspects of one or more components described with reference to FIG. **1**, such as one or more processors **170**.

(154) Storage **1140** may be configured to store data that is generated, processed, stored, or otherwise used by the system **1105**. In some cases, the storage **1140** may include one or more HDDs, one or more SSDs, or both. In some examples, the storage **1140** may be an example of a single database, a distributed database, multiple distributed databases, a data store, a data lake, or an emergency backup database. In some examples, the storage **1140** may be an example of one or more components described with reference to FIG. **1**, such as one or more network disks **180**.

(155) For example, the audit authorization component **1120** may be configured as or otherwise support a means for receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The audit authorization component **1120** may be configured as or otherwise support a means for identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The audit authorization component **1120** may be configured as or otherwise support a means for identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The audit authorization component **1120** may be configured as or otherwise support a means for determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(156) By including or configuring the audit authorization component **1120** in accordance with examples as described herein, the system **1105** may support techniques for access authorization for audit information in a multi-tenancy DMS, which may provide one or more benefits such as, for example, improved data management, compliance, efficiency, and security, among other possibilities.

(157) FIG. **12** illustrates a flowchart showing a method **1200** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The operations of the method **1200** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1200** may be performed by a DMS as described with reference to FIGS. **1** through **11**. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

(158) At **1205**, the method may include receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The operations of **1205** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1205** may be performed by a request component **1025** as described with reference to FIG. **10**.

(159) At **1210**, the method may include identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The operations of **1210** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1210** may be performed by a context

information component **1030** as described with reference to FIG. **10**.

(160) At **1215**, the method may include identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The operations of **1215** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1215** may be performed by an authorization information component **1035** as described with reference to FIG. **10**.

(161) At **1220**, the method may include determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity. The operations of **1220** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1220** may be performed by an access authorization component **1040** as described with reference to FIG. **10**.

(162) FIG. **13** illustrates a flowchart showing a method **1300** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The operations of the method **1300** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1300** may be performed by a DMS as described with reference to FIGS. **1** through **11**. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

(163) At **1305**, the method may include obtaining audit information associated with an entity, where entries within the audit information are associated with respective events including respective actions performed by the entity on respective objects. The operations of **1305** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1305** may be performed by an audit information component **1045** as described with reference to FIG. **10**.

(164) At **1310**, the method may include receiving, by a DMS, a request to provide a first user of the DMS with the audit information associated with the entity of the DMS, wherein the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The operations of **1310** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1310** may be performed by a request component **1025** as described with reference to FIG. **10**.

(165) At **1315**, the method may include identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The operations of **1315** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1315** may be performed by a context information component **1030** as described with reference to FIG. **10**.

(166) At **1320**, the method may include identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The operations of **1320** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1320** may be performed by an authorization information component **1035** as described with reference to FIG. **10**.

(167) At **1325**, the method may include determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity, where determining whether to output the audit information associated with the entity is further based on whether the tenant associated with

the request has access to audit information associated with the respective objects. The operations of **1325** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1325** may be performed by an access authorization component **1040** as described with reference to FIG. **10**.

(168) FIG. **14** illustrates a flowchart showing a method **1400** that supports access authorization for audit information in a multi-tenancy DMS in accordance with aspects of the present disclosure. The operations of the method **1400** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1400** may be performed by a DMS as described with reference to FIGS. **1** through **11**. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

(169) At **1405**, the method may include receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS. The operations of **1405** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1405** may be performed by a request component **1025** as described with reference to FIG. **10**.

(170) At **1410**, the method may include identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants. The operations of **1410** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1410** may be performed by a context information component **1030** as described with reference to FIG. **10**.

(171) At **1415**, the method may include identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants. The operations of **1415** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1415** may be performed by an authorization information component **1035** as described with reference to FIG. **10**.

(172) At **1420**, the method may include determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity. The operations of **1420** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1420** may be performed by an access authorization component **1040** as described with reference to FIG. **10**.

(173) At **1425**, the method may include receiving, from the first user as part of the log-in session associated with the tenant, a second request to view second audit information associated with one or more users of one or more other tenants, where the one or more other tenants include one or more subtenants of the tenant within a tenant hierarchy associated with the set of multiple tenants. The operations of **1425** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1425** may be performed by a request component **1025** as described with reference to FIG. **10**.

(174) At **1430**, the method may include filtering, per tenant or group of tenants and in response to the second request, a database of audit information stored by the DMS. The operations of **1430** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1430** may be performed by an audit information filtering component **1060** as described with reference to FIG. **10**.

(175) At **1435**, the method may include outputting, to the first user via a user interface in response

to the second request, the second audit information associated with the one or more users of the one or more other tenants based on the filtering and the one or more other tenants including the one or more subtenants of the tenant within the tenant hierarchy. The operations of **1435** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1435** may be performed by an audit information filtering component **1060** as described with reference to FIG. **10**.

(176) A method is described. The method may include receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS, identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants, identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants, and determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(177) An apparatus is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to receive, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS, identify, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants, identify, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants, and determine whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(178) Another apparatus is described. The apparatus may include means for receiving, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS, means for identifying, in response to the request, context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants, means for identifying, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants, and means for determining whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(179) A non-transitory computer-readable medium storing code is described. The code may include instructions executable by a processor to receive, by a DMS, a request to provide a first user of the DMS with audit information associated with an entity of the DMS, where the DMS includes a set of multiple computing objects, and where the DMS is operable to provide protection for data sources associated with a set of multiple tenants of the DMS, identify, in response to the request,

context information for a log-in session associated with the request, where the context information for the log-in session includes an ID of a tenant associated with the request, the tenant included in the set of multiple tenants, identify, based on the ID of the tenant associated with the request, authorization information associated with the tenant, where the authorization information indicates that the tenant has access to a set of entities within a hierarchy associated with the set of multiple tenants, and determine whether to output, to the first user in response to the request, the audit information associated with the entity based on a hierarchical relationship, within the hierarchy, between the tenant and the entity.

(180) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for obtaining the audit information associated with the entity, where entries within the audit information may be associated with respective events including respective actions performed by the entity on respective objects, and where determining whether to output the audit information associated with the entity may be further based on whether the tenant associated with the request may have access to audit information associated with the respective objects.

(181) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for obtaining the audit information associated with the entity, where entries within the audit information may be associated with respective events including respective actions performed by respective subjects on the entity, and where determining whether to output the audit information associated with the entity may be further based on whether the tenant associated with the request may have access to audit information associated with the respective subjects.

(182) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the authorization information indicates that the tenant may have access to an authentication domain within the hierarchy, the hierarchy including an authentication hierarchy associated with the set of multiple tenants and the hierarchical relationship indicates a relationship between the entity and the authentication domain in the authentication hierarchy.

(183) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for outputting, to the first user in response to the request, the audit information associated with the entity based on the hierarchical relationship indicating that the entity may be below the authentication domain within the authentication hierarchy.

(184) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for refraining from outputting, to the first user in response to the request, the audit information associated with the entity based on the hierarchical relationship indicating that the entity may be not below the authentication domain within the authentication hierarchy.

(185) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the authentication domain includes a node within the authentication hierarchy, the node above a single user of a set of multiple users of the DMS in the authentication hierarchy.

(186) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the authentication domain includes a node within the authentication hierarchy, the node above a user group of the DMS in the authentication hierarchy.

(187) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the authorization information indicates a set of one or more objects to which the tenant associated with the request may have access within the hierarchy, the hierarchy including an object hierarchy associated with a set of multiple objects of the DMS and the hierarchical relationship indicates a relationship between the entity and the set of one or more objects in the object hierarchy.

(188) Some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein may further include operations, features, means, or instructions for identifying an audit event for a third user of the DMS, identifying, in response to identifying the audit event, second context information for a second log-in session associated with the third user, where the second context information for the second log-in session includes a second ID of a second tenant associated with the third user, the second tenant included in the set of multiple tenants, and storing, in a database, second audit information associated with the audit event and the second ID of the second tenant associated with the third user.

(189) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for monitoring one or more operations performed by the third user of the DMS based on the third user initiating the second log-in session, where identifying the audit event includes identifying an operation of the one or more operations.

(190) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the first user as part of the log-in session associated with the tenant, a second request to view second audit information associated with one or more users of one or more other tenants, where the one or more other tenants include one or more subtenants of the tenant within a tenant hierarchy associated with the set of multiple tenants, filtering, per tenant or group of tenants and in response to the second request, a database of audit information stored by the DMS, and outputting, to the first user via a user interface in response to the second request, the second audit information associated with the one or more users of the one or more other tenants based on the filtering and the one or more other tenants including one or more subtenants of the tenant within the tenant hierarchy.

(191) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, filtering the database of audit information per tenant or group of tenants may include operations, features, means, or instructions for selecting the second audit information from among the database of audit information based on the second audit information including the ID of the tenant associated with the request, where remaining audit information in the database includes IDs of other tenants included in the set of multiple tenants.

(192) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first user initiates a second log-in session with the DMS after the log-in session, and second context information for the second log-in session associated with the first user includes a second ID of a second tenant included in the set of multiple tenants, the second tenant different than the tenant associated with the request.

(193) It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, aspects from two or more of the methods may be combined.

(194) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

(195) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar

components having the same first reference label irrespective of the second reference label.

(196) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(197) The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(198) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Further, a system as used herein may be a collection of devices, a single device, or aspects within a single device.

(199) Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(200) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, EEPROM) compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers.

Combinations of the above are also included within the scope of computer-readable media.

(201) The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method, comprising: storing, by a data management system, audit information that indicates information associated with one or more events performed by a second user on a computing object of the data management system, wherein the audit information further indicates respective identifiers of the second user and the computing object based at least in part on the second user being a subject of the one or more events and the computing object being an object of the one or more events, and wherein the data management system comprises a plurality of computing objects and is operable to provide protection for data sources associated with a plurality of tenants of the data management system; receiving, by the data management system, a request to provide a first user of the data management system with the audit information associated with the one or more events performed by the second user on the computing object, wherein the first user is associated with a first tenant of the plurality of tenants of the data management system, and wherein the second user is associated with a second tenant of the plurality of tenants of the data management system, identifying, in response to the request, context information for a log-in session associated with the request, wherein the context information for the log-in session comprises an identifier of the first tenant associated with the first user and the request, wherein the second tenant of the second user associated with the audit information is different than the first tenant associated with the first user and the request; identifying, based at least in part on the identifier of the first tenant associated with the first user and the request, authorization information associated with the first tenant, wherein the authorization information indicates: a set of one or more computing objects to which the first tenant associated with the first user and the request has access within an object hierarchy associated with the plurality of computing objects of the data management system in accordance with a hierarchical relationship, within the object hierarchy, between the first tenant and the set of one or more computing objects; and a set of one or more second users to which the first tenant associated with the first user and the request has access within an authentication hierarchy associated with the plurality of tenants and a plurality of users of the data management system in accordance with a second hierarchical relationship, within the authentication hierarchy, between the first tenant and an authentication domain comprising the set of one or more second users; and determining to output, to the first user associated with the first tenant and in response to the request, the audit information associated with the one or more events performed on the computing object by the second user associated with the second tenant based at least in part on the set of one or more computing objects indicated via the authorization information comprising at least the computing object associated with the audit information and further based at least in part on the set of one or more second users indicated via the authorization information comprising at least the second user associated with the audit information.

2. The method of claim 1, wherein determining to output the audit information is further based at least in part on whether the first tenant associated with the first user and the request has access to the audit information associated with the one or more events performed on the computing object in accordance with the authorization information.

3. The method of claim 1, further comprising: obtaining the audit information, wherein entries within the audit information are associated with the one or more events comprising respective actions performed by the second user on the computing object, and wherein determining to output

the audit information associated is further based at least in part on whether the first tenant associated with the first user and the request has access to the audit information associated with the second user.

4. The method of claim 1, wherein the authorization information indicates that the first tenant has access to the authentication domain within the authentication hierarchy.

5. The method of claim 4, further comprising: outputting, to the first user in response to the request, the audit information based at least in part on the second hierarchical relationship indicating that the second user is below the authentication domain within the authentication hierarchy.

6. The method of claim 4, wherein the authentication domain comprises a node within the authentication hierarchy, the node above a single user of the plurality of users of the data management system in the authentication hierarchy.

7. The method of claim 4, wherein the authentication domain comprises a node within the authentication hierarchy, the node above a user group of the data management system in the authentication hierarchy.

8. The method of claim 1, further comprising: identifying an audit event for a third user of the data management system; identifying, in response to identifying the audit event, second context information for a second log-in session associated with the third user, wherein the second context information for the second log-in session comprises a second identifier of a third tenant associated with the third user, the third tenant included in the plurality of tenants; and storing, in a database, second audit information associated with the audit event and the second identifier of the third tenant associated with the third user.

9. The method of claim 8, further comprising: monitoring one or more operations performed by the third user of the data management system based at least in part on the third user initiating the second log-in session, wherein identifying the audit event comprises identifying an operation of the one or more operations.

10. The method of claim 1, further comprising: receiving, from the first user as part of the log-in session associated with the first tenant, a second request to view second audit information associated with one or more second users of one or more other tenants, wherein the one or more other tenants comprise one or more subtenants of the first tenant within a tenant hierarchy associated with the plurality of tenants; filtering, per tenant or group of tenants and in response to the second request, a database of audit information stored by the data management system; and outputting, to the first user via a user interface in response to the second request, the second audit information associated with the one or more second users of the one or more other tenants based at least in part on the filtering and the one or more other tenants comprising the one or more subtenants of the first tenant within the tenant hierarchy.

11. The method of claim 10, wherein filtering the database of audit information per tenant or group of tenants comprises: selecting the second audit information from among the database of audit information based at least in part on the second audit information comprising the identifier of the first tenant associated with the request, wherein remaining audit information in the database comprises identifiers of other tenants included in the plurality of tenants.

12. The method of claim 1, wherein the first user initiates a second log-in session with the data management system after the log-in session, and second context information for the second log-in session associated with the first user comprises a second identifier of a third tenant included in the plurality of tenants, the third tenant different than the first tenant associated with the request.

13. An apparatus, comprising: a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to: store, by a data management system, audit information that indicates information associated with one or more events performed by a second user on a computing object of the data management system, wherein the audit information further indicates respective identifiers of the second user and the computing object based at least in part on the second user being a subject of the one or more events and the

computing object being an object of the one or more events, and wherein the data management system comprises a plurality of computing objects and is operable to provide protection for data sources associated with a plurality of tenants of the data management system; receive, by the data management system, a request to provide a first user of the data management system with the audit information associated with the one or more events performed by the second user on the computing object, wherein the first user is associated with a first tenant of the plurality of tenants of the data management system, and wherein the second user is associated with a second tenant of the plurality of tenants of the data management system; identify, in response to the request, context information for a log-in session associated with the request, wherein the context information for the log-in session comprises an identifier of the first tenant associated with the first user and the request, wherein the second tenant of the second user associated with the audit information is different than the first tenant associated with the first user and the request; identify, based at least in part on the identifier of the first tenant associated with the first user and the request, authorization information associated with the first tenant, wherein the authorization information indicates; a set of one or more computing objects to which the first tenant associated with the first user and the request has access within an object hierarchy associated with the plurality of computing objects of the data management system in accordance with a hierarchical relationship, within the object hierarchy, between the first tenant and the set of one or more computing objects; and a set of one or more second users to which the first tenant associated with the first user and the request has access within an authentication hierarchy associated with the plurality of tenants and a plurality of users of the data management system in accordance with a second hierarchical relationship, within the authentication hierarchy, between the first tenant and an authentication domain comprising the set of one or more second users; and determine to output, to the first user associated with the first tenant and in response to the request, the audit information associated with the one or more events performed on the computing object by the second user associated with the second tenant based at least in part on the set of one or more computing objects indicated via the authorization information comprising at least the computing object associated with the audit information and further based at least in part on the set of one or more second users indicated via the authorization information comprising at least the second user associated with the audit information.

14. The apparatus of claim 13, wherein the instructions are executable by the processor to cause the apparatus to determine to output the audit information further based at least in part on whether the first tenant associated with the first user and the request has access to the audit information associated with the one or more events performed on the computing object in accordance with the authorization information.

15. The apparatus of claim 13, wherein the instructions are further executable by the processor to cause the apparatus to: obtain the audit information, wherein entries within the audit information are associated with the one or more events comprising respective actions performed by the second user on the computing object, and wherein determining to output the audit information associated is further based at least in part on whether the first tenant associated with the first user and the request has access to the audit information associated with the second user.

16. The apparatus of claim 13, wherein the instructions are further executable by the processor to cause the apparatus to: identify an audit event for a third user of the data management system; identify, in response to identifying the audit event, second context information for a second log-in session associated with the third user, wherein the second context information for the second log-in session comprises a second identifier of a third tenant associated with the third user, the third tenant included in the plurality of tenants; and store, in a database, second audit information associated with the audit event and the second identifier of the third tenant associated with the third user.

17. A non-transitory computer-readable medium storing code, the code comprising instructions executable by a processor to: store, by a data management system, audit information that indicates information associated with one or more events performed by a second user on a computing object

of the data management system, wherein the audit information further indicates respective identifiers of the second user and the computing object based at least in part on the second user being a subject of the one or more events and the computing object being an object of the one or more events, and wherein the data management system comprises a plurality of computing objects and is operable to provide protection for data sources associated with a plurality of tenants of the data management system; receive, by the data management system, a request to provide a first user of the data management system with the audit information associated with the one or more events performed by the second user on the computing object, wherein the first user is associated with a first tenant of the plurality of tenants of the data management system, and wherein the second user is associated with a second tenant of the plurality of tenants of the data management system; identify, in response to the request, context information for a log-in session associated with the request, wherein the context information for the log-in session comprises an identifier of the first tenant associated with the first user and the request, wherein the second tenant of the second user associated with the audit information is different than the first tenant associated with the first user and the request; identify, based at least in part on the identifier of the first tenant associated with the first user and the request, authorization information associated with the first tenant, wherein the authorization information indicates: a set of one or more computing objects to which the first tenant associated with the first user and the request has access within an object hierarchy associated with the plurality of computing objects of the data management system in accordance with a hierarchical relationship, within the object hierarchy, between the first tenant and the set of one or more computing objects; and a set of one or more second users to which the first tenant associated with the first user and the request has access within an authentication hierarchy associated with the plurality of tenants and a plurality of users of the data management system in accordance with a second hierarchical relationship, within the authentication hierarchy, between the first tenant and an authentication domain comprising the set of one or more second users; and determine to output, to the first user associated with the first tenant and in response to the request, the audit information associated with the one or more events performed on the computing object by the second user associated with the second tenant based at least in part on the set of one or more computing objects indicated via the authorization information comprising at least the computing object associated with the audit information and further based at least in part on the set of one or more second users indicated via the authorization information comprising at least the second user associated with the audit information.

18. The non-transitory computer-readable medium of claim 17, wherein the instructions are executable by a processor to determine to output the audit information further based at least in part on whether the first tenant associated with the first user and the request has access to the audit information associated with the one or more events performed on the computing object in accordance with the authorization information.
